K-Bound: When Is Label-Free Adaptation Knowable? Helpful, Harmful, and Unknowable Regimes Under Distribution Shift

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***Abstract*—Test-time adaptation (TTA) can recover accuracy from unlabeled target data, but the same update can silently degrade a deployed model under a different shift. We study the decision that precedes adaptation: given a frozen model f_0, a candidate adapted model f_a, and label-free deployment evidence Z, should the system *adapt*, *freeze*, or *abstain*? We define the target adaptation benefit as $\Delta = R_T(f_0) - R_T(f_a)$ and characterize when its sign is recoverable without target labels.**

Our central result is an exact benefit-sign frontier over a declared calibration-drift class. On the disagreement region, the sign of Δ is identifiable if and only if an observable margin exceeds the admitted drift budget, $|M| > \beta$. Below this frontier, target worlds with indistinguishable label-free evidence can require different deployment actions, so a valid rule must retain an abstention region. On the identifiable side, a calibrated interval $\widehat{\Delta} \pm \varepsilon$ yields a finite-sample adapt/freeze/abstain certificate controlling marginal false-adapt and false-freeze probabilities under stated coverage, exchangeability, and risk-alignment assumptions.

We instantiate the framework as Knowability-Guided Adaptation (KGA), a mechanism-agnostic wrapper around Tent, EATA, and SAR. Across nine benchmark tracks, the empirical behavior follows the regime geometry: mixed and detectable stress regimes yield confirmed wins over both always-adapt and always-freeze; one-sided natural shifts primarily yield no-harm behavior by matching the better fixed policy; and weak or non-transferable evidence produces abstention or diagnostic failure rather than a promoted win. K-Bound is therefore a validity layer for deciding whether an adaptation update is supportable, not a universal accuracy booster.

***Index Terms*—test-time adaptation, distribution shift, selective prediction, identifiability, label-free risk estimation**

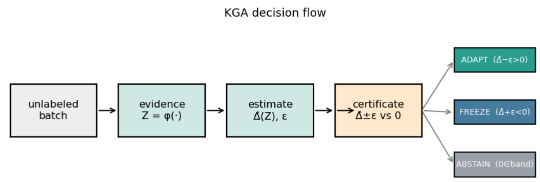


Fig. 1. **K-Bound previews the decision.** From an unlabeled batch under shift, the certificate estimates the adaptation benefit $\widehat{\Delta}$ with a finite-sample radius ε and returns ADAPT (knowably helpful), FREEZE (knowably harmful), or ABSTAIN (sign unidentifiable), with false-adapt/false-freeze $\leq \alpha$.

I. Introduction

Test-time adaptation (TTA) updates a model on the unlabeled test stream and, under favorable distribution shift, recovers accuracy. Under unfavorable shift, the same objective silently degrades it, and with no target labels the system usually cannot tell which case it faces. The decision that matters therefore comes *before* the update: given a frozen baseline f_0 and an adapted candidate f_a, choose **adapt** when adaptation is knowably helpful, **freeze** when knowably harmful, or **abstain** when label-free evidence cannot identify the sign of the benefit. The field's default assumption is that adaptation should be made robust enough to always run; we replace it with an explicit, certified decision about whether to run it at all.

Some cases are information-theoretically unknowable. It is established that label-free quantities cannot decide adaptation without an assumption on the shift: identifying OOD accuracy is as hard as identifying the optimal predictor [13], and the optimal predict-versus-abstain rule under a shift budget has a known form [14]. We sharpen this landscape, for the benefit sign, into a computable criterion: whether adaptation helps or hurts is label-free identifiable *if and only if* an observable margin exceeds the calibration-drift budget on the disagreement region, i.e. $|M| > \beta$ (Theorem 2). On the unidentifiable side, two target worlds carry identical label-free evidence but opposite benefit, so no label-free rule can be correct in both (Theorem 1) and abstention is forced. KGA (Algorithm 1), a benefit estimator plus a conformal radius wrapped around any adapter (Tent, EATA, SAR), operationalizes the frontier as a finite-sample certificate holding false-adapt and false-freeze at level α (Theorem 3).

The evidence follows the regime map in Table I. *Identifiable mixed* harmful+helpful regimes supply the **beats-both** policy wins, including a pre-registered head-to-head over protocol-matched POEM-style and AETTA-style baselines. Natural shifts primarily test safety: a reconciled result may tie the better fixed policy, while a provisional result remains outside the headline until its saved summary and canonical replay agree. K-Bound does *not* claim universal improvement: when adaptation is already safe, a valid certificate can only tie always-adapt. It provides a safety layer for TTA rather than an accuracy boost.

a) Plain-language takeaway.: If label-free evidence is strong enough relative to declared drift ($|M| > \beta$), the system can certify ADAPT or FREEZE; if not, it should ABSTAIN. On single natural-shift datasets this mostly acts as safety or abstention. Beats-both wins require a detectable mixed regime; a constructed pooled mixture is reported as such, not as a claim of single-dataset natural-shift dominance.

b) Contributions.:

1) **Problem framing.** Label-free TTA as a certified *adapt/freeze/abstain* decision on the sign of the adaptation benefit.
2) **Exact frontier.** The benefit sign is label-free identifiable *iff* $|M| > \beta$; below the frontier, matched evidence forces abstention (Theorems 1, 2; §IV).
3) **Finite-sample certificate.** KGA controls the unconditional false-adapt/false-freeze rate at level α under stated coverage (Theorem 3); existing label-free estimators are its β=0 face.
4) **Drop-in method.** KGA wraps Tent/EATA/SAR without changing their adaptation objectives (Algorithm 1; §V).
5) **Uniform empirical accounting.** A nine-track panel reports the same metrics, split logic, and evidence tier for every benchmark; only locked or reconciled results are promoted to empirical claims (§VII).

Table note. The uniform panel below is the sole empirical index; diagnostic variants and historical runs are not additional evidence for the headline claims.

TABLE I

Regime map: expected behavior vs. result. EACH REGIME, AN EXAMPLE DATASET, THE BEHAVIOR THE FRONTIER PREDICTS FOR A VALID CERTIFICATE, AND WHAT WAS MEASURED. "NO-HARM" = MATCHES THE BETTER FIXED POLICY AND BEATS THE WORSE, WITHOUT BEATING BOTH; "BEATS BOTH" = LOWER REGRET THAN ALWAYS-ADAPT *AND* ALWAYS-FREEZE. THE EVIDENCE TIER IN TABLE XIV STATES WHETHER THAT POINT VERDICT IS CI-CONFIRMED, RECONCILED, PROVISIONAL, OR DIAGNOSTIC.

Regime	Example	Expected	Result
Helpful-dominated	Camelyon17 OOD	tie always-adapt	reconciled no-harm
Harmful-dominated	RxRx1; iWildCam/Office-Home OOF summaries	tie always-freeze, beat always-adapt	no-harm / provisional
Domain-gen. natural	PACS (4 LODO splits)	no-harm or abstain	three safety tracks; one null
Mixed + detectable	CIFAR-10-C stress, ImageNet-C SAR	beat both fixed policies	CI-confirmed (both)
Weak evidence	ImageNet-R, CIFAR-10.1	abstain / conservative	diagnostic; no promoted win
Constructed mixture	three-source OOF stream	route per condition, beat both	supported routing result, not transfer

II. Related Work

The closest literature improves test-time updates, monitors adaptation failures, estimates target performance without labels, or characterizes abstention under shift. K-Bound differs in the decision object: it certifies the sign of the frozen-versus-adapted risk difference before the candidate state is committed.

A. Test-Time Adaptation

Tent [1], SHOT [2], TTT [3], EATA [4], SAR [5], CoTTA [6], and MEMO [7] improve adaptation through entropy minimization, filtering, regularization, sharpness-aware updates, or temporal stabilization (survey: [66]). These methods primarily answer *how* to adapt. Even when the update rule is more stable, the deployment system generally still commits to adaptation rather than certifying that the candidate has positive target benefit.

B. Guarded and Monitored Adaptation

POEM [11], sequential risk monitors [9], [10], PeTTA [73], and sample-level filters such as DeYO [74] guard an adaptation stream during or after updating. These approaches are complementary to K-Bound. They typically monitor a proxy or intervene in one failure direction, whereas K-Bound poses a pre-commitment three-way decision over the benefit sign and explicitly represents an unknowable region.

C. Label-Free Performance Estimation

ATC [13], agreement-on-the-line [16], [15], AETTA [18], and related confidence-style estimators [65], [64] estimate target accuracy or error without target labels. Estimating $R_T(f)$ is not identical to certifying $\mathrm{sign}(R_T(f_0) - R_T(f_a))$: a deployment controller needs the sign of a paired risk difference and must account for uncertainty around zero. K-Bound uses this difference as the estimand and abstains when the available evidence does not identify its sign.

D. Impossibility, Abstention, and Risk Control

General label-free target-risk identification requires assumptions on the shift [13], [21], [22], [25]. Kalai and Kanade [14] characterize a related abstain-versus-predict decision under a shift budget, while selective prediction [37] abstains on individual labels. K-Bound specializes this landscape to adaptation benefit, localizes the comparison to the disagreement region, and derives an adapt/freeze/abstain certificate. Its empirical uncertainty layer connects to split conformal prediction, conformal under shift, RCPS, and Learn-then-Test [67], [69], [68], [71], [72].

E. Surviving Gap

Existing work provides stronger adapters, reactive monitors, label-free accuracy estimates, and general abstention theory. The remaining gap is a pre-commitment rule for the sign of adaptation benefit that (i) distinguishes helpful, harmful, and unsupported decisions, (ii) states the assumptions under which the decision is valid, and (iii) controls marginal false adaptation under a calibrated uncertainty model. K-Bound addresses this gap with the population frontier and its finite-sample realization, KGA.

III. Problem Formulation and Validity

A. Deployment Setting and Adaptation Benefit

Source data follow $P_S(X,Y)$ and may be labeled during training and development. Deployment data follow $P_T(X,Y)$, but only target inputs and model outputs are available when the decision is made. A frozen predictor f_0 serves as the safe fallback, and an adapter $\mathcal{A}$ proposes a temporary candidate $f_a = \mathcal{A}(f_0, X_{1:m})$ from an unlabeled target batch. For loss ℓ, the target risk and adaptation benefit are

$$R_T(f) = \mathbb{E}_{P_T}[\ell(f(X),Y)], \qquad \Delta = R_T(f_0) - R_T(f_a). \quad (1)$$

Thus $\Delta > 0$ favors committing the candidate, $\Delta < 0$ favors retaining the frozen state, and $\Delta = 0$ makes the two actions risk-equivalent. Target labels reveal Δ only during offline auditing; they are not available to the deployment rule.

B. Target-Label-Free Evidence and Decision Semantics

The controller observes a label-free evidence vector $Z = \phi(X_{1:m}, f_0, f_a)$ containing quantities such as prediction entropy, confidence, class-balance statistics, frozen-versus-adapted disagreement, output-distribution drift, and update magnitudes. The deployment rule returns

$$g(Z) \in \{\text{ADAPT}, \text{FREEZE}, \text{ABSTAIN}\}.$$

ADAPT commits f_a; FREEZE rejects the proposal and retains f_0 (or the latest certified state in continual operation); and ABSTAIN declines to change the model, serves the frozen prediction, and may request more evidence or human review. Abstention therefore applies to the update decision, not to prediction itself.

C. What "Label-Free" Means

K-Bound is target-label-free rather than supervision-free. Source labels and a held-out development set may be used to fit the benefit estimator and calibrate its uncertainty radius. No target-test label may influence the adapter configuration, evidence map, benefit estimator, radius, action threshold, or model selection. Target-test labels are opened only after all decisions are frozen, for reporting accuracy, regret, and false-adapt events.

D. Assumptions and Validity Scope

The theory and the finite-sample certificate rely on distinct assumptions. *Risk alignment* determines whether the evidence can identify or bound the benefit sign over the declared target class. *Coverage* determines whether the empirical interval contains the true benefit at the stated rate. *Exchangeability* (or an explicit shift correction) connects calibration residuals to deployment residuals. The adapter, evidence schema, estimator, and calibration protocol must be fixed before held-out scoring. If these conditions cannot be justified, the theorem does not license a commitment; the operational fallback is freeze or abstain.

TABLE II
Data access by stage. Target-test labels are forbidden until offline audit.

Stage	Source labels	Dev labels	Target-test
Source training	yes	optional	no
Benefit/radius calibration	optional	yes	no
Deployment decision	no	frozen artifacts only	no
Offline evaluation	no	no	yes, after dec

TABLE III
Assumptions, their role, and the safe response when they are unsupported.

Condition	Role	If unsupported
Risk-aligned Z	Benefit sign is recoverable over the declared class	abstain; do not interpret a point estimate as a certificate
Coverage of $\widehat{\Delta} \pm \varepsilon$	Controls marginal false-adapt/freeze	recalibrate, widen the radius, or freeze
Exchangeable/corrected residuals	Transfers calibration to deployment	use shift-aware calibration or abstain
Bounded loss/benefit	Supports concentration and finite-sample analysis	clip/redefine the loss or state no bound
Frozen protocol	Prevents target-test leakage and selection bias	rerun with a locked protocol

E. Notation and Observability

F. Multiclass and Regression Scope

The binary 0–1 loss case gives the cleanest frontier because, on $D = \{x : f_0(x) \neq f_a(x)\}$, exactly one predictor is correct. For multiclass 0–1 loss,

$$\Delta = P_T(D)(p_a - p_0), \qquad p_j = P_T(f_j(X) = Y \mid D),$$

so the decision becomes an ordinal comparison of the two disagreement-region accuracies. For squared-loss regression, the risk difference reduces to a disagreement-local moment. The main theorem uses a split-observable decomposition $\operatorname{sign}(\Delta) = \operatorname{sign}(M + \gamma)$; complete multiclass and regression statements remain in the appendix.

Source $P_S(X,Y)$ has labels at train/validation; target $P_T(X,Y)$ exposes only $P_T(X)$ at test time. The central question is when $\operatorname{sign}\Delta$ is recoverable from label-free evidence Z alone. Risk alignment (Definition 1) governs identifiability; coverage governs the finite-sample certificate (Theorem 3). When Z is not risk-aligned, Theorem 1 forces abstention.

G. Formal Setup

Source law $P_S(X,Y)$ supplies labels at train/validation; target law $P_T(X,Y)$ exposes only $P_T(X)$ at deployment. Fix loss ℓ, frozen predictor f_0, candidate f_a, and target measure

TABLE IV

Core quantities. The population drift budget β and empirical radius ε are related but not interchangeable.

Symbol	Meaning	Available at deployment?
f_0, f_a	frozen and candidate models	yes
Δ	true target adaptation benefit	no
D	disagreement region	yes
M	observable population margin	through label-free evidence
γ	realized calibration drift	no
β	declared bound on $\|\gamma\|$	supplied pre-deployment
Z	label-free evidence vector	yes
$\widehat{\Delta}$	estimated adaptation benefit	yes
ε	calibrated residual radius	fixed pre-deployment
α	marginal error level	chosen pre-deployment

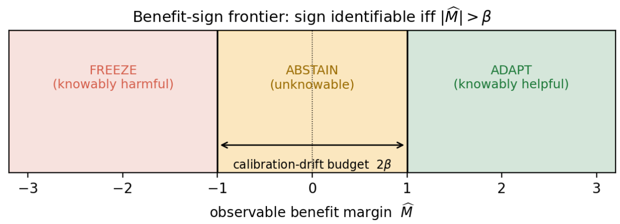


Fig. 2. **The benefit-sign frontier.** The sign of the adaptation benefit is identifiable from label-free evidence iff the observable margin $|\widehat{M}|$ exceeds the calibration-drift budget β. Outside the band KGA adapts (knowably helpful) or freezes (knowably harmful); inside the band (the unknowable region), abstention is information-theoretically forced. ($\widehat{M}$ is the finite-sample estimate of the population margin M of Theorem 2.)

μ_T on inputs. Write target risk $R_T(f) = \mathbb{E}_{P_T}[\ell(f(X), Y)]$ and the *adaptation benefit*

$$\Delta \;:=\; R_T(f_0) - R_T(f_a) \qquad (\Delta > 0 \;\Leftrightarrow\; \text{adapting helps}).$$

The *disagreement region* $D := \{x : f_0(x) \neq f_a(x)\}$ is observable from (f_0, f_a).

Assumption 1 (Deployment setup)**.** Unless stated otherwise: binary label $Y \in \{0,1\}$, 0/1 loss ℓ, and $\mu_T(D) > 0$. Fix a source-calibrated score $s : \mathcal{X} \to [0,1]$ estimating the candidate's *pointwise correctness* (e.g. a Platt-scaled confidence that f_a is right at x), and write the target pointwise correctness $\eta_a(x) := \mathbb{P}_T\big(f_a(X){=}Y \mid X{=}x\big)$ on D, so that $\mathbb{E}_{\mu_T}[\eta_a \mid D] = \bar{a}$. Define the *observable margin*

$$M \;:=\; \mathbb{E}_{\mu_T}[s(X) \mid D] - \tfrac{1}{2},$$

the *calibration drift* (unobserved at deploy)

$$\gamma \;:=\; \mathbb{E}_{\mu_T}[\eta_a(X) - s(X) \mid D],$$

and label-free evidence $Z = \phi(X_{1:m}, f_0, f_a)$ for any measurable ϕ using only unlabeled batches and model outputs (entropy, confidence, drift, disagreement, update norms). A *label-free rule* is any measurable map $g : \mathcal{Z} \to \{\text{ADAPT}, \text{FREEZE}, \text{ABSTAIN}\}$ that does not use target labels on the deployment batch.

Definition 1 (Risk alignment)**.** Evidence Z is *risk-aligned on a class $\mathcal{P}$* if no two laws $P_T^1, P_T^2 \in \mathcal{P}$ with $\operatorname{sign}\Delta_1 \neq \operatorname{sign}\Delta_2$ satisfy $\operatorname{Law}(Z \mid P_T^1) = \operatorname{Law}(Z \mid P_T^2)$.

Definition 2 (Regimes)**.** At evidence z: *knowably helpful* if the population frontier certifies $\Delta > 0$ (ADAPT); *knowably harmful* if it certifies $\Delta < 0$ (FREEZE); *unknowable* if the same Z-law is compatible with both signs (ABSTAIN).

Remark 1 (Four quantities). Do not confuse M (observable margin, deploy-time), γ (realized drift, latent), β (declared bound $|\gamma| \leq \beta$ on the deployment class), and ε (finite-sample radius from held-out residuals). The radius ε operationalizes coverage; it is *not* an estimate of β.

Symbol	Meaning	At deploy?	Role
M	observable evidence margin	yes	evidence toward adapt/freeze
γ	realized calibration drift	no	can reverse the benefit sign
β	declared bound on $\|\gamma\|$	externally supplied	population frontier
ε	residual quantile radius	estimated pre-deploy	finite-sample certificate

What "label-free" means. No target/test labels at deploy; labeled source and held-out validation may calibrate $\widehat{\Delta}$ and ε. Evidence Z uses only unlabeled batches, predictions, entropy, drift, and update statistics. The method is **target-label-free**, not label-free of all supervision: labels may be used only in the source/validation calibration split, never in the deployment batch or held-out test decision.

K-Bound in one sentence. The true adaptation-benefit sign is recoverable only when label-free evidence exceeds the uncertainty created by calibration drift; otherwise the correct action is abstention.

For a supplied drift budget $\beta \geq 0$, the *declared deployment class* is

$$\mathcal{C}_\beta \;:=\; \big\{P_T : \; |\gamma(P_T)| \leq \beta\big\}.$$

Identifiability statements below are *over* $\mathcal{C}_\beta$, not over arbitrary shifts.

IV. Theory

A. Roadmap

The theoretical spine has three layers. First, the risk difference reduces to the region where the frozen and candidate predictions disagree. Second, matched label-free evidence can support incompatible benefit signs, which establishes an abstention requirement. Third, the decomposition $\operatorname{sign}(\Delta) = \operatorname{sign}(M{+}\gamma)$ yields the population frontier, while a calibrated interval around $\widehat{\Delta}$ yields the finite-sample decision rule.

Lemma 1 (Disagreement-region reduction)**.** *Under Assumption 1, with $\bar{a} := \mathbb{P}_T(f_a{=}Y \mid D)$,*

$$\Delta \;=\; 2\,\mu_T(D)\big(\bar{a}{-}\tfrac{1}{2}\big), \qquad \textit{so} \qquad \operatorname{sign}\Delta \;=\; \operatorname{sign}(\bar{a}{-}\tfrac{1}{2}) \;=\; \operatorname{sign}(M{+}\gamma$$

The identity $\operatorname{sign}\Delta = \operatorname{sign}(M+\gamma)$ *holds for* every *target law; it does not assume* $|\gamma| \le \beta$.

Proof. On D exactly one of f_0, f_a is correct under 0/1 loss, so the benefit equals $2\mu_T(D)(\bar{a} - \frac{1}{2})$. By definition of M and γ, $\bar{a} - \frac{1}{2} = \mathbb{E}_{\mu_T|D}[\eta_a - s] = M + \gamma$. □

Theorem 1 (Matched evidence forces abstention)**.** *Fix* $\alpha \in (0, \frac{1}{2})$ *and* $\beta \ge 0$. *There exist* $P_T^1, P_T^2 \in \mathcal{C}_\beta$ *such that* $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$, $\operatorname{sign}\Delta_1 = -\operatorname{sign}\Delta_2 \ne 0$, *and* $|M(P_T^1)| < \beta$. *Consequently, any label-free rule* g *with* $\mathbb{P}[g = \textsc{adapt}, \Delta \le 0] \le \alpha$ *and* $\mathbb{P}[g = \textsc{freeze}, \Delta \ge 0] \le \alpha$ *must satisfy* $\mathbb{P}[g = \textsc{abstain}] \ge 1 - 2\alpha$ *on this shared evidence law.*

Proof. Take $|M| < \beta$ and choose $\gamma_1, \gamma_2 \in [-\beta, \beta]$ with $\operatorname{sign}(M + \gamma_1) \ne \operatorname{sign}(M + \gamma_2)$, both nonzero (possible because $|M| < \beta$; at the boundary $|M| = \beta$ the admissible drift $\gamma = -\operatorname{sign}(M)\beta$ instead gives $\Delta = 0$, still uncertifiable as a strict sign, so abstention persists on the closed region $|M| \le \beta$). Lemma 1 gives opposite benefit signs at the same observables $(M, \mu_T(D), Z\text{-law})$. A label-free rule sees identical $\mathrm{Law}(Z)$ in both worlds, so its action probabilities coincide; each committed action is a false adapt in one world or a false freeze in the other, forcing abstention with probability at least $1 - 2\alpha$. An explicit Gaussian witness appears in Appendix A (Theorem 4). □

Theorem 2 (Exact benefit-sign frontier)**.** *Under Assumption 1, fix the observables* $(\mu_T, P_S, f_0, f_a, s)$ *so* M *is determined, and let* $\beta \ge 0$.

- (i) ***Reduction.*** *For every* P_T, $\operatorname{sign}\Delta = \operatorname{sign}(M+\gamma)$ *(Lemma 1).*
- (ii) ***Sufficiency.*** *If* $|M| > \beta$ *and* $P_T \in \mathcal{C}_\beta$, *then* $\operatorname{sign}\Delta = \operatorname{sign} M$.
- (iii) ***Necessity.*** *If* $|M| < \beta$, *there exist* $P_T^1, P_T^2 \in \mathcal{C}_\beta$ *with the same evidence law and opposite* nonzero *benefit signs; at* $|M| = \beta$ *the admissible drift* $\gamma = -\operatorname{sign}(M)\beta$ *yields* $\Delta = 0$, *so the sign is still undetermined. Hence* $\operatorname{sign}\Delta$ *is not a function of* $\mathrm{Law}(Z)$ *over* $\mathcal{C}_\beta$ *whenever* $|M| \le \beta$.
- (iv) ***Iff.*** $\operatorname{sign}\Delta$ *is label-free identifiable over* $\mathcal{C}_\beta$ *if and only if* $|M| > \beta$.

The population rule $M > \beta \Rightarrow \textsc{adapt}$, $M < -\beta \Rightarrow \textsc{freeze}$, $|M| \le \beta \Rightarrow \textsc{abstain}$ *is the maximal sound certificate on* $\mathcal{C}_\beta$.

Proof. Part (i) is Lemma 1. For (ii), if $M > \beta$ and $|\gamma| \le \beta$, then $M + \gamma > 0$, so $\operatorname{sign}\Delta = \operatorname{sign}(M + \gamma) = +1$; the negative case is symmetric. For (iii), pick $\gamma_\pm \in [-\beta, \beta]$ with $\operatorname{sign}(M + \gamma_+) \ne \operatorname{sign}(M + \gamma_-)$, both nonzero, when $|M| < \beta$; Lemma 1 yields matched Z and opposite signs. At $|M| = \beta$ the admissible $\gamma = -\operatorname{sign}(M)\beta$ gives $\Delta = 0$, still precluding a determined nonzero sign. Part (iv) combines (ii) and (iii). Maximality follows from (iii) and Theorem 1. □

Region	Meaning	Action
$M > \beta$	allowed drift cannot flip the positive sign	ADAPT
$M < -\beta$	allowed drift cannot flip the negative sign	FREEZE
$\lvert M\rvert \le \beta$	drift can flip the sign	ABSTAIN

Theorem 3 (Finite-sample adapt/freeze/abstain certificate)**.** *Under Assumption 1, suppose calibration and deployment pairs are exchangeable and the benefit estimator satisfies marginal coverage*

$$\mathbb{P}\big[\,|\widehat{\Delta} - \Delta| \le \varepsilon\,\big] \;\ge\; 1 - \alpha.$$

Define

$$g(Z) = \begin{cases} \textit{ADAPT} & \widehat{\Delta} - \varepsilon > 0, \\ \textit{FREEZE} & \widehat{\Delta} + \varepsilon < 0, \\ \textit{ABSTAIN} & \textit{otherwise.} \end{cases}$$

Then, marginally over the calibration randomness,

$$\mathbb{P}\big[g(Z) = \textit{ADAPT},\, \Delta \le 0\big] \le \alpha, \qquad \mathbb{P}\big[g(Z) = \textit{FREEZE},\, \Delta \ge 0\big] \le \alpha.$$

Proof. On the event $\{|\widehat{\Delta} - \Delta| \le \varepsilon\}$, the condition $\widehat{\Delta} - \varepsilon > 0$ implies $\Delta > 0$, so a false adapt requires the complement of this event, which has probability at most α. The freeze case is symmetric. □

Remark 2 (Marginal vs. conditional error)*.* Theorem 3 bounds the *marginal, unconditional* false-adapt and false-freeze probabilities ($\mathrm{FA_u}$). It does *not* bound the conditional rate $\mathrm{FA_c} = \mathbb{P}[\Delta \le 0 \mid \textsc{adapt}]$, which we report empirically. Risk alignment (Definition 1) is required for the *frontier* identifiability claim; coverage alone certifies the *interval* rule above.

Corollary 1 (Unjustified assumptions $\Rightarrow$ abstain)**.** *If exchangeability, coverage, or risk alignment cannot be justified on the deployment batch, Theorem 3 does not apply. Committing to* ADAPT *without a valid positive certified lower bound is not protected at level* α*; the conservative action is* ABSTAIN *or* FREEZE *(cf. Theorem 1).*

B. How the Results Fit Together

disagreement-region reduction $\to$ matched-evidence ambiguity $\to$ **Thm. 1** (abstention necessity)
$\to$ split-observable decomposition $\operatorname{sign}\Delta = \operatorname{sign}(M+\gamma)$ $\to$ **Thm. 2** ($|M| > \beta$ frontier)
$\to$ coverage condition $\to$ **Thm. 3** (finite-sample certificate)

C. Extensions and Deferred Results

One-bit orientation (with the weakest one-bit class characterized unconditionally as a finite family of dominance polytopes; Appendix B, Theorem 8), minimax order-optimality on the identifiable side, multicandidate family-wise routing, anytime-valid streaming, and the Wave-4 dichotomies (exact three-world minimax constant, margin computability, multiclass capacity, regression bracketing) are proved and validated in Appendix A and the long manuscript (`kbound.tex`). They strengthen the framework but are not required to understand the main K-Bound result. The streaming form is demonstrated on real data: on the native iWildCam stream (raw stream artifact offline; replay pending), the betting e-process detects continual-Tent collapse with anytime false-adapt 0, and the gate ties the oracle policy (pre-registered). **[DRAFT TODO: Streaming window/gate figures (35,370-image run) withheld: no saved result artifact was found on T9; rerun `gapclose_wave5/win_hunt_D_anytime_stream.py` to produce and lock one.]**

D. Claim Scope

The population theorem is conditional on the declared class $\mathcal{C}_\beta$, and the finite-sample guarantee is conditional on valid interval coverage. The theorem controls the marginal probabilities $\mathbb{P}(\textsc{adapt}, \Delta \leq 0)$ and $\mathbb{P}(\textsc{freeze}, \Delta \geq 0)$; it does not automatically control the conditional error among committed actions. The framework does not claim that β is identifiable from unlabeled deployment data, that the empirical radius ε estimates β, or that calibration coverage transfers to arbitrary unseen domains without an assumption or correction.

V. KGA Method

A. System Overview

KGA turns the frontier certificate into a deployable rule. It is the only headline method—a decision wrapper around a fixed adapter, not a new adaptation objective. We fit the benefit estimator and its radius *out of fold*, so the data used to fit $\widehat{\Delta}$ and the data used to calibrate its residual are disjoint for every point. On the per-cell stress grids (CIFAR-10-C and the synthetic grids), for each cell i a gradient-boosted $\widehat{\Delta}_{-i}$ is trained on the other $N-1$ cells and evaluated on cell i, and ε is the $(1-\alpha)$ empirical quantile of the leave-one-out residuals $\{|\widehat{\Delta}_{-i}(Z_i) - \Delta_i|\}_{i=1}^{N}$—leave-one-out (jackknife) cross-conformal, so no cell's residual uses an estimator that saw it. On the natural-shift protocols (Camelyon17, iWildCam, Office-Home) the split is across domains: $\widehat{\Delta}$ and ε are fit once on a dev split and the held-out test domain is scored a single time—ordinary split-conformal. At deploy we extract Z from an unlabeled batch and return adapt, freeze, or abstain (Algorithm 1). Labels enter only through the calibration split; the deployment batch and held-out test scorer never use target labels for the adapter, ε, evidence, or rule selection. The coverage guarantee is split-dependent. The natural-shift protocols use a clean dev/test split: *split-conformal* coverage $\mathbb{P}[|\widehat{\Delta} - \Delta| \leq \varepsilon] \geq 1-\alpha$, exact in finite samples under exchangeability of the calibration and test cells. The per-cell grids set ε to the $(1-\alpha)$ empirical quantile of the leave-one-out residuals, so the *calibration-set* coverage is $\geq 1-\alpha$ by construction (realized 0.898 at nominal 0.90, α=0.10); this is the standard jackknife radius, which is distribution-free only asymptotically under estimator stability; its formal finite-sample counterpart is the jackknife+ of Barber et al. (2021) at level $1-2\alpha$, which we note but do not implement. The decision rule's false-adapt guarantee is stated under the coverage hypothesis (§V), which the empirical coverage meets. This construction sits in a known landscape: weighted conformal corrects coverage under known covariate shift [67], adaptive conformal tracks coverage online [69], and beyond exchangeability the coverage gap is bounded by the total-variation drift of the calibration pairs [68]—the population counterpart of our declared budget β. The certificate's marginal false-adapt control is risk control in the RCPS/Learn-then-Test lineage [71], [72], specialized to the adapt/freeze/abstain loss; PAC prediction sets give set-valued analogues under covariate shift [70].

B. Evidence Map and Benefit Estimator

The evidence map uses only unlabeled batches and model outputs, and is instantiated as two label-free panels matched to the two calibration regimes below. On the controlled stress grids (CIFAR-10-C, ImageNet-C, PACS) the base panel is an 11-dimensional vector computed from the frozen softmax p_0 and the adapted softmax p_a on the adaptation batch: the mean predictive entropy and mean confidence of f_0 and of f_a; the normalized marginal (predicted-class) entropy of each; the entropy and marginal-balance drops between them; the fraction of adapted predictions above confidence 0.9 (a collapse indicator); the KL divergence between the adapted and frozen predicted-class marginals; and the ℓ_2 norm of the adapter's parameter update. On the natural-shift protocols (Camelyon17, iWildCam, Office-Home, RxRx1) a 16-dimensional panel adds distribution- and disagreement-oriented statistics computed from the frozen logits: entropy and confidence quantiles; the frozen/adapted prediction-disagreement rate; the entropy gap; a free-energy source→target shift; a batch-vs-running BatchNorm-statistic KL; an ATC-style threshold-accuracy estimate (its threshold fixed on *source* labels only, never target labels); and a source→target confidence drop. The source-calibrated correctness score s of §IV is operationalized by the adapted-confidence and ATC-style features, and the observable margin M is their disagreement-region summary.

The benefit model predicts $\widehat{\Delta} = h_\theta(Z)$ from labeled development conditions; h_θ is a gradient-boosted regression tree (scikit-learn `GradientBoostingRegressor`, squared error, 250 trees, depth 2, learning rate 0.05, subsample 0.8, seed 0), refit *per dataset and per candidate adapter* rather than shared, so the estimator itself implies no cross-dataset transfer. The complete schema for both panels—exact per-feature formulas and families—is given in Appendix A (Tables XXI and XXII), with per-adapter update hyperparameters in Table XXIII.

C. Out-of-Fold Uncertainty Calibration

For calibration condition i, an estimator that did not train on i produces $\widehat{\Delta}_{-i}(Z_i)$ and residual

$$r_i = |\widehat{\Delta}_{-i}(Z_i) - \Delta_i|.$$

TABLE V

LABEL-FREE EVIDENCE *ACTUALLY COMPUTED* BY KGA. THE 11-DIM BASE PANEL DRIVES THE STRESS GRIDS; THE NATURAL-SHIFT PROTOCOLS USE A 16-DIM PANEL THAT ADDS THE ROWS MARKED *(NAT.)*. EXACT FORMULAS AND PER-TRACK SUBSETS ARE IN THE CONFIGURATION FILES.

Family	Statistics
Frozen output Z_0	mean entropy, mean confidence, normalized marginal-class entropy; entropy/confidence quantiles *(nat.)*
Adapted output Z_a	mean entropy, mean confidence, normalized marginal entropy, frac. confidence > 0.9
Change Z_{change}	entropy drop, marginal-balance drop, entropy gap
Disagreement	frozen/adapted prediction-disagreement rate *(nat.)*
Distribution drift	adapted-vs-frozen marginal KL; free-energy shift, BN-statistic KL, ATC/confidence drop *(nat.)*
Update behavior	ℓ_2 norm of the adapter parameter update

Algorithm 1 KGA: Knowability-Guided Adaptation

Require: Frozen f_0; adapter $\mathcal{A}$; batch $X_{1:m}$; $\widehat{\Delta}$, ε, α
Ensure: Decision $g \in \{\text{ADAPT}, \text{FREEZE}, \text{ABSTAIN}\}$

$f_a \leftarrow \mathcal{A}(f_0, X_{1:m})$
$Z \leftarrow \phi(X_{1:m}, f_0, f_a)$
$\widehat{\Delta} \leftarrow \widehat{\Delta}(Z)$; $L \leftarrow \widehat{\Delta} - \varepsilon$; $U \leftarrow \widehat{\Delta} + \varepsilon$
if $L > 0$ **then**
  $g \leftarrow$ ADAPT
else if $U < 0$ **then**
  $g \leftarrow$ FREEZE
else
  $g \leftarrow$ ABSTAIN
end if
return g

The radius is a pre-specified upper residual quantile. On clean development/test splits, ordinary split conformal provides finite-sample marginal coverage under exchangeability. On leave-one-cell-out stress grids, the reported radius is a jackknife-style empirical residual quantile; its calibration-set coverage does not by itself imply the same finite-sample guarantee as split conformal or jackknife+. We therefore report the calibration design and the guarantee level separately for each track.

On the good event $|\widehat{\Delta} - \Delta| \leq \varepsilon$, committing adapt only when $L > 0$ implies $\Delta > 0$ (Theorem 3). Assumptions: exchangeable calibration pairs, bounded benefits, and risk-aligned Z; no target labels at deploy. Overhead: one forward pass of f_0 and f_a plus lightweight $\widehat{\Delta}$ evaluation.

D. Population Frontier versus Empirical Certificate

KGA's commit boundary is not a swept hyperparameter, but the population frontier and the empirical radius must be kept distinct. The *population* frontier of Theorem 2 (over the declared class $\mathcal{C}_\beta$, $\operatorname{sign}\Delta$ is identifiable iff the disagreement margin $|M|$ exceeds the supplied budget β) fixes the *required* budget at population level; the drift realization $\gamma \leq \beta$ is unknown and not observed at deploy.

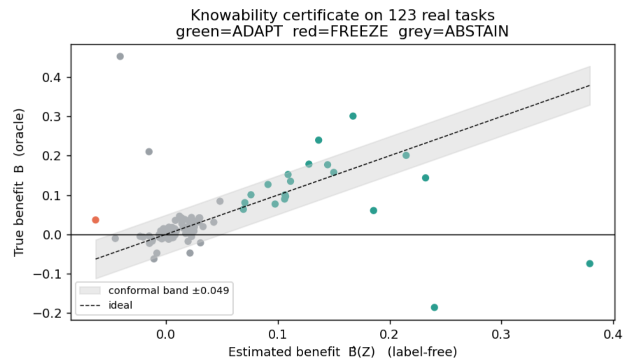


Fig. 3. **The certificate rule in one picture.** KGA estimates the label-free benefit $\widehat{\Delta}(Z)$ and forms an interval $\widehat{\Delta} \pm \varepsilon$: if the interval is entirely positive it adapts, if entirely negative it freezes, and otherwise it abstains. The vertical axis here uses true benefit only for offline audit; deployment decisions use only unlabeled evidence and the pre-calibrated radius.

KGA does not observe γ; it commits under a *conservative empirical radius* ε, the $(1-\alpha)$ residual quantile on the calibration split, fixed there and *never* selected on target-test data. The two are related but not identical: ε is a finite-sample, assumption-dependent surrogate that controls false-adapt under the stated exchangeability and risk-alignment assumptions, and is not the frontier itself. Confidence-thresholding (ATC) and agreement-trend selection (AGL) fit a scalar against a labeled or held-out signal and predict *accuracy*; KGA certifies the benefit *sign* with a boundary *forced* by the identifiability budget, and, unlike any thresholded estimator, ABSTAINS on the region where Theorem 1 *proves* no label-free rule can decide. The impossibility construction, not a tuned cutoff, is what separates KGA from "estimate accuracy, then threshold."

E. Selecting and Auditing the Drift Budget

The drift budget β is part of the declared deployment class, not a target-test-tuned hyperparameter. It may be justified from historical domain calibration gaps, a documented domain assumption, or an explicit stress envelope. Sensitivity analysis should report how decisions change over a pre-specified range of budgets. A smaller β asserts stronger calibration transfer and increases commitment; a larger β widens the abstention region. When no credible upper bound is available, the population frontier does not provide a robust sign certificate. Reporting $\beta = 0$ is then only a zero-drift plug-in analysis, not an assumption-free or conservative guarantee.

F. Deployment Semantics and Failure-Safe Behavior

The candidate update is evaluated in a temporary or shadow state. ADAPT commits the candidate; FREEZE rolls it back; and ABSTAIN also retains the frozen state but records that the evidence was insufficient rather than certifiably harmful. The implementation should automatically freeze or abstain when evidence is missing, the

TABLE VI
KGA CONTROLLER OVERHEAD (*ADDED* COST; RESNET-18/CIFAR, MEASURED ON CPU VIA `SCRIPTS/COST_PROFILE.PY`). THE ADAPTATION STEP ITSELF IS UNCHANGED BY KGA.

Component	Cost
Evidence assembly (per batch)	0.073 ms
Benefit-model evaluation (per decision)	0.125 ms
Controller added latency	**0.20 ms**
Rollback model copy (memory)	44.8 MB
Benefit model size	195 KB

batch is too small, a checkpoint or adapter configuration differs from calibration, the evidence lies outside calibration support, numerical values are invalid, or no valid radius is available. In the reference implementation (`kbound_pkg`), the gate is applied per batch (`KGA.decide_from_batch`); on ABSTAIN or FREEZE the adaptation optimizer scales the candidate update by `abstain_scale`=0, i.e. the update is skipped and the frozen state f_0 is served—this is the rollback mechanism. The failure-safe defaults instantiate the diagnostics of Table XIX: abstain when the batch is smaller than the calibration batch size, when Z lies outside calibration support, when a checkpoint or adapter-configuration hash differs from the calibrated one, or when the radius or evidence is missing or numerically invalid. In continual operation the last certified state replaces f_0 as the fallback.

G. Computational Cost

KGA may not avoid the cost of *proposing* an update because f_a must be evaluated before commitment. Its deployment value is preventing harmful state changes and their accumulation. The cost analysis must therefore separate candidate-adaptation time, evidence extraction, benefit-model evaluation, model-copy/rollback memory, and total latency. Table VI reports the *added* controller cost, since f_a must be evaluated regardless: on ResNet-18/CIFAR the decision layer adds 0.20 ms per batch decision (evidence assembly 0.073 ms + benefit-model evaluation 0.125 ms) and one model copy (44.8 MB) for rollback; the benefit model itself is 195 KB. The adaptation step (forward+backward) dominates end-to-end latency and is unchanged by KGA.

VI. EXPERIMENTAL SETUP

A. Research Questions

The experiments address seven questions: **RQ1** whether the measured commit transition follows the theoretical frontier; **RQ2** whether the interval controls marginal false adaptation; **RQ3** whether KGA beats both fixed policies in mixed detectable regimes; **RQ4** whether it prevents damage on held-out natural shifts; **RQ5** whether it improves over simple and neighboring label-free gates; **RQ6** which evidence and calibration choices matter; and **RQ7** what computational overhead the controller adds.

B. Datasets and Shift Taxonomy

The evaluation separates controlled corruption/composition stress, large-scale corruption, natural institutional or geographic shifts, domain generalization, and low-margin diagnostic transfers. This taxonomy matters because a router can beat both fixed policies only when the optimal action varies across conditions and that variation is detectable from label-free evidence. The nine tracks draw on CIFAR-10-C [51], ImageNet [50], the WILDS Camelyon17, iWildCam, and RxRx1 shifts [53], [54], [55], Office-Home [56], PACS via DomainBed [57], and CIFAR-10.1 [59]; models build on RobustBench checkpoints [58] with PyTorch [60] and scikit-learn [61].

C. Models, Adapters, and Leakage Control

For every track, the source checkpoint, candidate adapter, update mode, learning rate, number of steps, batch construction, adapted parameter subset, and random seeds must be declared before held-out scoring. Adapter selection, evidence selection, benefit-model fitting, and residual calibration use development data only. All choices are frozen before the target test cells are scored once; target labels are opened afterward solely for offline audit. The calibration and scoring configuration is uniform across tracks except where noted in Table VIII.

Exact per-adapter update settings (optimizer, steps, learning rate, and the mild/aggressive operating points) are tabulated per track in Appendix A, Table XXIII.

D. Baselines

We distinguish fixed policies (always-freeze and always-adapt), an offline per-condition oracle, simple label-free gates (confidence, entropy, drift/KL, and ATC-style), KGA without its radius, and neighboring methods or ports inspired by POEM and AETTA. Each comparison must state whether the baseline is an official implementation, a faithful reimplementation, or a protocol-matched port.

E. Metrics and Statistical Testing

The primary decision metrics are regret-to-oracle, marginal false-adapt $\mathrm{FA_u} = \mathbb{P}(\textsc{adapt}, \Delta \leq 0)$, conditional false-adapt $\mathrm{FA_c} = \mathbb{P}(\Delta \leq 0 \mid \textsc{adapt})$, adapt rate, and decision coverage $\mathbb{P}(\textsc{adapt or freeze})$. Accuracy or AUROC is reported as the task metric. The theorem controls $\mathrm{FA_u}$ under its assumptions; $\mathrm{FA_c}$ is descriptive. A beats-both verdict requires lower regret than both fixed policies under the declared statistical criterion and $\mathrm{FA_u} \leq \alpha$. Paired bootstrap confidence intervals, their resampling unit, seed aggregation, and Holm correction are declared per comparison family.

F. Evidence-Status Policy

Locked results trace to raw cells/seeds and a frozen aggregate; **reconciled** results correct a prior interpretation using raw records; **provisional** summaries require canonical replay; and **diagnostic** runs are complete but

TABLE VII

DATASET ROLES IN THE EVALUATION. THE EXPECTED REGIME IS A HYPOTHESIS FIXED BY THE PROTOCOL, NOT A CONCLUSION INFERRED FROM THE FINAL SCORE.

Track	Shift type	Candidate role	Evaluation purpose
CIFAR-10-C stress	corruption/composition/update stress	Tent/EATA/SAR	mixed detectable decision benchmark
ImageNet-C noise	large-scale corruption	Tent/EATA/SAR	scale and candidate dependence
Camelyon17	hospital/institution shift	EATA	helpful-dominated natural safety
iWildCam	geographic camera shift	Tent	harmful-dominated natural safety
Office-Home	visual domain/style shift	SAR	held-out domain safety
RxRx1	experimental/laboratory shift	online adapter	harmful-dominated natural safety
PACS	domain generalization	fixed candidate	breadth and null behavior
ImageNet-R	rendition shift	candidate panel	weak-evidence diagnostic
CIFAR-10.1	natural resampling	TTA candidates	low-margin transfer diagnostic

TABLE VIII

PER-TRACK CALIBRATION AND SCORING CONFIGURATION (α=0.10 THROUGHOUT; BENEFIT ESTIMATOR IDENTICAL ACROSS TRACKS, §V). BACKBONES: RESNET-18 (CIFAR), RESNET-50 (IMAGENET-C); THE NATURAL-SHIFT BACKBONES FOLLOW THE WILDS/DOMAINBED DEFAULTS PER TRACK.

Track	Calibration unit	Radius rule	Test unit
CIFAR-10-C stress	grid cell (leave-one-cell-out)	LOO jackknife $q_{0.9}$	5×432 cells
ImageNet-C	grid cell (leave-one-cell-out)	LOO jackknife $q_{0.9}$	27 cells
PACS (LODO)	source-domain cells	LOO jackknife $q_{0.9}$	108 cells/target
Camelyon17, iWildCam, Office-Home, RxRx1	dev seeds $\{0, 1\}$	split-conformal $q_{0.9}$	test seeds $\{2, 3, 4\}$

deliberately not promoted. Only locked and reconciled rows support the empirical headline.

VII. RESULTS

The results are organized by regime rather than by dataset prestige. Controlled mixed regimes test policy improvement, one-sided natural shifts test no-harm behavior, and low-margin transfers test whether the method declines unsupported claims.

A. Synthetic validation of the frontier

Before the real-data tracks we validate the theory directly on synthetic ground truth, where the observable margin M, the unobserved drift $\gamma \sim \mathrm{U}(-\beta, \beta)$, and hence the true benefit sign $\operatorname{sign}\Delta = \operatorname{sign}(M + \gamma)$ are all controlled. Evidence is fed to the exact deployed rule (leave-one-out gradient-boosted $\widehat{\Delta}$ with a split-conformal radius, §V); nothing is hand-tuned. Three predictions hold (Fig. 5; α=0.10). **(i) Recovery:** $\widehat{\Delta}$ tracks Δ at 90.0% empirical coverage—exactly the target $1 - \alpha$—and the radius self-calibrates to $\varepsilon \approx \beta$ (the unseen drift *is* the residual). **(ii) Frontier** (Thm 2): the commit rate jumps from 6.5% for $|M| < \beta$ to 96.7% for $|M| > \beta$—a transition *at* the predicted boundary—and the committed sign is correct 99.3% of the time. **(iii) Certificate** (Thm 3): the unconditional false-adapt rate stays under budget for every $\beta \in \{0.05, \ldots, 0.20\}$ ($\mathrm{FA_u} \leq 0.014 \ll \alpha$). The frontier is not imposed—it emerges because the conformal radius equals the drift the evidence cannot see.

candidate	policy	mean acc	regret↓	$\mathrm{FA_c}$	coverage	beats both?
Tent	always-adapt	0.786	0.0079	—	1.00	
	always-freeze	0.671	0.1241	—	0.00	
	K-Bound	**0.793**	**0.0016**	**0**	0.67	**yes**
EATA	always-adapt	0.798	0.0033	—	1.00	
	always-freeze	0.671	0.1314	—	0.00	
	K-Bound	**0.801**	**0.0013**	**0**	0.63	**yes**
SAR	always-adapt	0.806	0.0003	—	1.00	
	always-freeze	0.671	0.1405	—	0.00	
	K-Bound	0.806	0.0015	0	—	ties adapt

TABLE IX

CIFAR-10-C STRESS GRID (432 CONDITIONS/METHOD, FIVE SEEDS, α=0.1, PROTOCOL A).

B. CIFAR-10-C stress grid (5 seeds)

The per-corruption suite is helpful-dominated; the *stress* grid crosses severity, batch size, composition, and update aggressiveness (432 conditions/method, ResNet-18; pre-registered as Protocol A; seeds 0–4). Harmful base rates are 34% (Tent), 27% (EATA), ≈10% (SAR, mean over seeds). KGA **beats both** trivial policies for Tent and EATA at 0% false-adapt; it ties collapse-resistant SAR (Table IX). Among harmful cells KGA adapts 0 and freezes them; among helpful cells it adapts almost all; on marginal cells it abstains. For both Tent and EATA, the regret gaps to *both* fixed policies are positive with 95% bootstrap CIs excluding zero, Holm-corrected over the comparison family, at false-adapt 0/2160.[1] The mixing-ratio Pareto (Fig. 6) shows KGA on the regret-vs-harmful-fraction front, strictly beating both once the harmful fraction exceeds $p^{\star} \approx 0.1$ (Tent/SAR) or 0 (EATA).

C. Decision-baseline comparison: a finite-sample false-adapt guarantee

KGA's claim goes beyond "better than always-adapt/always-freeze": it is better than *simple label-free decision rules* under the same locked protocol. On the

[1]Design-based intervals, resampling the pre-registered mixing-ratio distribution of the stress-grid design (Protocol A; the 11-point Pareto over harmful fraction, N_{boot}=5000): Tent +0.019 [0.011, 0.026] vs. always-adapt, +0.080 [0.052, 0.107] vs. always-freeze; EATA +0.015 [0.009, 0.021] and +0.075 [0.049, 0.100]. A conventional paired per-condition bootstrap over the 432 realized conditions (`scripts/percondition_bootstrap.py`) returns the same verdict; helpful-dominated SAR beats always-freeze but not always-adapt, all at $\mathrm{FA_u}$=0.

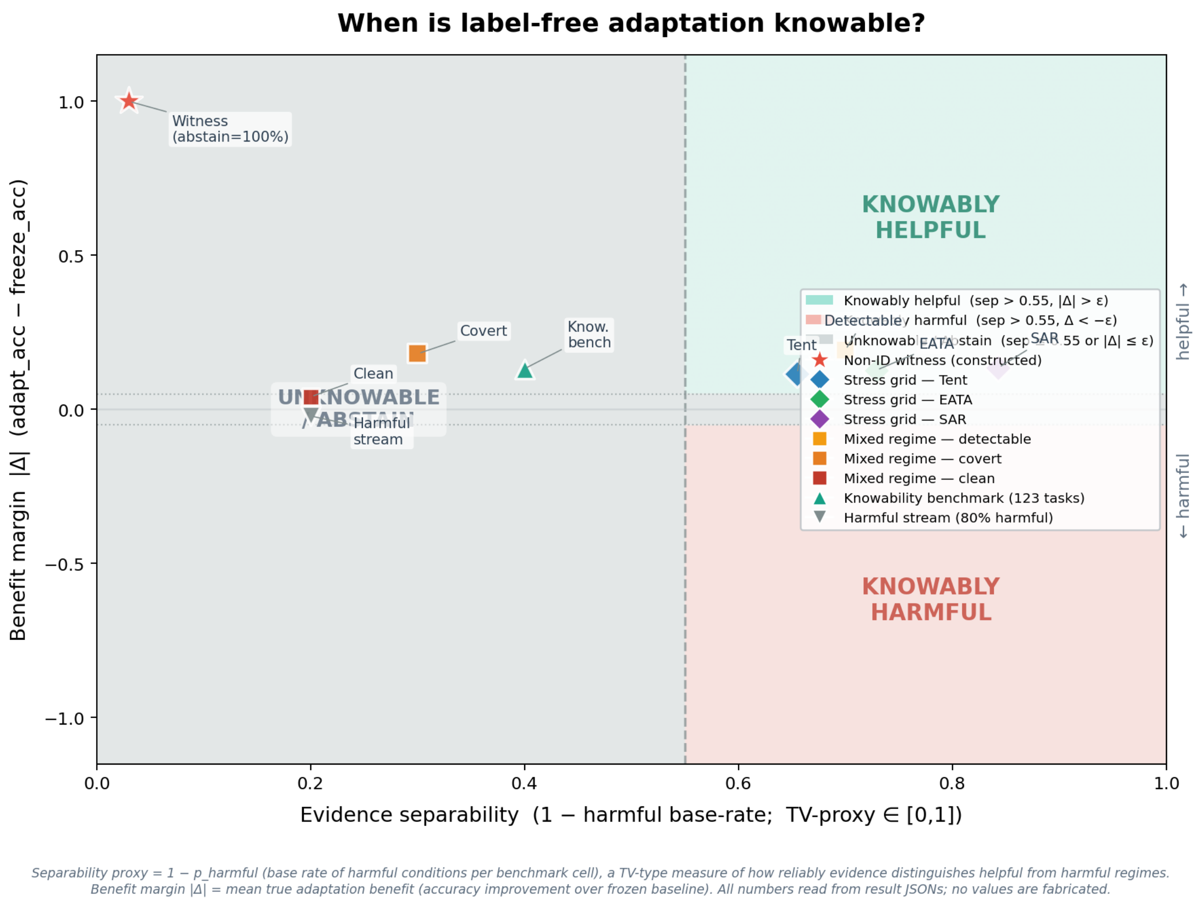


Fig. 4. **Regime geometry for K-Bound.** The important axis is not dataset name but where the deployment sits relative to the frontier. Helpful- or harmful-dominated regimes usually produce no-harm ties with the better fixed policy; low-margin regimes force abstention; mixed and detectable regimes are where a valid adapt/freeze/abstain router can beat both fixed policies. This is the visual map behind Table I and the experiment split below.

CIFAR-10-C stress grid (Tent candidate, identical cells), we score six rules from the same evidence (Table X): a confidence gate (adapt iff adaptation raised mean confidence), an entropy gate (adapt iff it lowered entropy—Tent's own objective), a drift/KL threshold, an ATC-style accuracy-prediction gate, KGA *without* its conformal radius (adapt iff $\widehat{\Delta} > 0$), and the full certificate. Only the certificate provides a finite-sample false-adapt *guarantee*, and it attains zero observed $\mathrm{FA_u}$ on this grid while staying near-oracle on regret (the radius-free variant keeps $\mathrm{FA_u}$=0.049<α empirically, but with no such guarantee). The confidence/entropy gates false-adapt on ~74% of harmful cells; the drift threshold, tuned to the budget, degenerates to always-freeze. The KGA-without-radius row isolates the radius: removing it lowers regret slightly (0.0017→0.0004) but raises $\mathrm{FA_u}$ from 0 to 0.049 (to 0.141 on harmful cells): the radius is what turns a good benefit *estimate* into a false-adapt *guarantee*. Reproduced by `scripts/gate_baseline_comparison.py` on the locked per-cell dump.

TABLE X
Decision-baseline comparison, CIFAR-10-C stress grid
(Tent candidate, n=432 cells, 149 harmful, α=0.1). Lower regret and lower $\mathrm{FA_u}$ are better. Only the conformal certificate keeps $\mathrm{FA_u}$=0 on the full grid and on harmful cells while staying near-oracle.

Decision rule	regret↓	$\mathrm{FA_u}$	$\mathrm{FA_c}$	adapt	cov.	$\mathrm{FA_u}$(harm)
confidence gate	0.0084	0.257	0.301	0.85	1.00	0.745
entropy gate	0.0086	0.255	0.304	0.84	1.00	0.738
drift/KL gate	0.1232	0.000	0.000	0.00	1.00	0.000†
ATC-style gate	0.0045	0.116	0.172	0.67	1.00	0.336
KGA (no radius)	0.0004	0.049	0.071	0.68	1.00	0.141
KGA (certificate)	0.0017	**0.000**	**0.000**	0.51	0.68	**0.000**

†Degenerate: the budget-tuned drift gate never adapts (adapt-rate 0), so its "0 false-adapt" is just always-freeze (regret 0.123).

D. *Mixed head-to-head vs. POEM-/AETTA-style baselines (pre-registered)*

The stress-grid wins above beat the *trivial* policies always-adapt and always-freeze. A harder question is whether KGA improves over strong protocol-matched label-free decision baselines inspired by POEM [11] and AETTA [18], on a *mixed* harmful+helpful

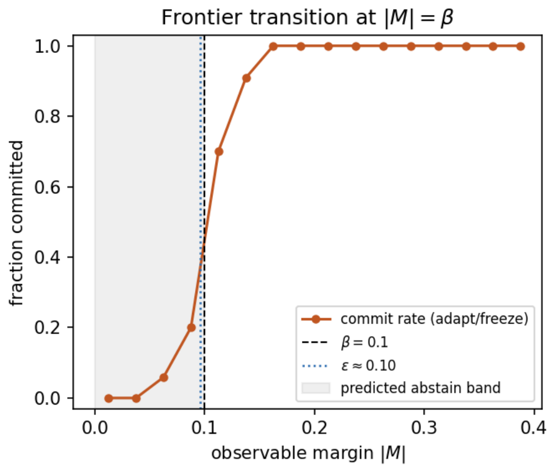


Fig. 5. **Measured benefit-sign frontier** (synthetic ground truth, exact deployed rule). The commit rate transitions sharply from abstention to adapt/freeze at $|M|=\beta$; the self-calibrated conformal radius ε (dotted) coincides with β. Companion panels—estimator recovery and $\mathrm{FA_u} \leq \alpha$ across β—in `fig_frontier_recovery.png`, `fig_frontier_fa_coverage.png`.

stream. We pre-registered the benchmark, metrics, and WIN/TIE/LOSE criterion *before* computing any number (`MIXED_BENCHMARK_PROTOCOL.md`). All six policies consume the *same* logged per-condition signals Z and benefit B from the locked CIFAR-10-C stress grid (Tent, 432 conditions/seed, five seeds; harmful fraction ≈33%, i.e. the always-adapt $\mathrm{FA_u}$ of Table XI); only the decision rule differs. The competitors are *protocol-matched POEM-style and AETTA-style baselines*: ports of the published protectors onto this protocol with documented simplifications (batch-summary entropy rather than per-sample streams). Official per-sample POEM and dropout-based AETTA reproduction remains a camera-ready faithfulness check.

Verdict: WIN. KGA lowers mean regret-to-oracle vs. both ported baselines with paired-bootstrap 95% CIs entirely below zero and Holm correction (α=0.05), at $\mathrm{FA_u}=0 \leq \alpha$ (Table XI). The ported POEM-/AETTA-style baselines false-adapt at ∼24–25% on the same stream, since they lack the certificate's marginal false-adapt control. Secondary pooled sets (Tent+EATA, EATA alone) yield the same WIN verdict (`experiments/kbound/results/mixed_headtohead_v1/`). An independent post-hoc recomputation (paired bootstrap 10^4, pre-registered) replicates all three configurations, with both regret-gap CIs below zero in each and $\mathrm{FA_{KGA}}=0$ versus 11–33% for the competitors (`research_lock/WIN_HUNT_v3_ARM_F_result.json`).

a) Baseline faithfulness.: The POEM and AETTA rows in Table XI are protocol-matched ports, not official reproductions. They are included to test whether KGA's conformal benefit-sign certificate improves over strong label-free committal gates under the same logged evidence,

TABLE XI

Pre-registered mixed head-to-head on the locked CIFAR-10-C stress stream. POEM/AETTA rows are protocol-matched style ports, not official reproductions. All policies use the same held-out cells, candidate adapter, logged label-free evidence, regret-to-oracle metric, and unconditional false-adapt metric. The comparison tests KGA against strong label-free decision gates under this locked protocol, not against full official POEM/AETTA implementations.

policy	mean regret↓	$\mathrm{FA_u}$	decisive rate	KGA gap	Holm beats?
always-adapt	0.0079	0.329	1.00	−0.0064	yes
always-freeze	0.1241	0.000	1.00	−0.1225	yes
POEM-style	0.0088	0.245	1.00	−0.0072 [−0.0088, −0.0057]	yes
AETTA-style	0.0073	0.240	1.00	−0.0058 [−0.0071, −0.0044]	yes
KGA	**0.0016**	**0.0000**	0.6847	—	**WIN**
oracle	0.0000	0.000	1.00	—	ceiling

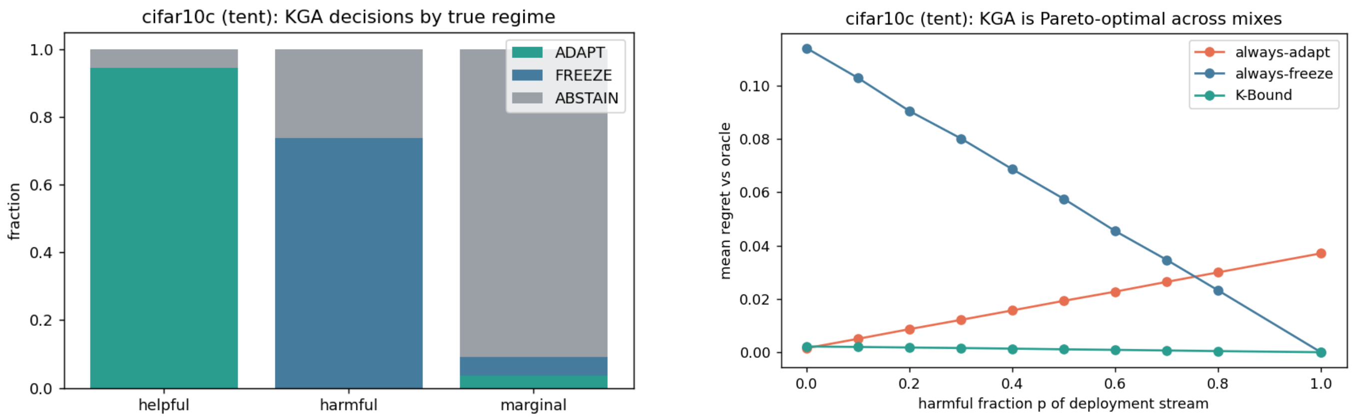
candidate adapter, held-out cells, regret metric, and false-adapt metric (Table XII). They do not claim to reproduce POEM's full per-sample betting process or AETTA's full dropout-based estimator. Official-code reproduction is left as an additional faithfulness check; the present claim is therefore limited to the locked protocol-matched comparison.

E. ImageNet-C SAR (mechanism-faithful)

On ImageNet-C noise (ResNet-50, 27 cells: three noise corruptions × severities $\{1, 3, 5\}$ × label regimes, seed 0, full-scale), SAR is harmful on 14.8% of cells under a mechanism-faithful re-implementation (SAM, recovery reset, EMA) at the learning rate shared across candidates. Always-adapt drops to 0.379 accuracy while KGA reaches 0.431, **beating both** trivial policies (regret 0.0108 vs 0.0625/0.0319). The gap to the better fixed policy (always-freeze) has a 95% paired-bootstrap CI of $[-0.032, -0.011]$ and survives Holm correction across the confirmatory wave, so ImageNet-C SAR is a *CI-confirmed* beats-both, not a point estimate. EATA stays helpful-dominated (KGA ties always-adapt); on Tent the harmful regime now dominates (63% of cells) and KGA abstains to always-freeze (Table XIII). Under the same logged conditions, an ATC-style confidence gate false-adapts on the harmful SAR cells—it adapts wherever confidence is high, which is exactly where SAR collapses—while KGA abstains on them, so the certificate beats not only the two fixed policies but a simple label-free decision gate under the same protocol. The result uses our mechanism-faithful SAR reimplementation at a shared learning rate; SAR's official gentler schedule is left as a robustness check, so the claim is about this harmful-SAR operating point rather than all SAR configurations.

F. Uniform nine-track panel and methodology

All benchmark tracks use the same decision target: the adaptation benefit $\Delta = R(f_0) - R(f_a)$, evaluated against the same per-condition oracle, always-adapt, and always-freeze policies. The candidate adapter is locked using development data. Evidence Z, the benefit estimator $\widehat{\Delta}$, and radius ε are then frozen before scoring the declared test cells; test labels are used only after decisions for offline



(a) decisions by true regime

(b) regret as harmful fraction changes

Fig. 6. **CIFAR-10-C stress grid is a decision result, not only an accuracy result.** KGA adapts on helpful cells, freezes harmful cells, abstains on marginal cells, and therefore lies on the regret front: it ties always-adapt at p=0 and beats both fixed policies once harmful mass is nontrivial.

TABLE XII
BASELINE FAITHFULNESS AUDIT FOR POEM/AETTA COMPARISONS.

Baseline	Original method needs	Our locked protocol uses	What is matched	What is not claimed
POEM-style	online entropy/betting monitor over adaptation stream	batch-summary entropy and logged per-condition evidence	same held-out cells, same candidate, same false-adapt/regret metrics	not official per-sample POEM reproduction
AETTA-style	dropout/disagreement-based label-free accuracy estimation	protocol-matched accuracy/benefit gate from logged evidence	same evidence budget, same decision task, same held-out stream	not official dropout-AETTA reproduction
KGA	benefit estimator + conformal radius	out-of-fold $\widehat{\Delta}$ and ε	same stream, same oracle/regret/$\mathrm{FA_u}$ metric	official method in this paper

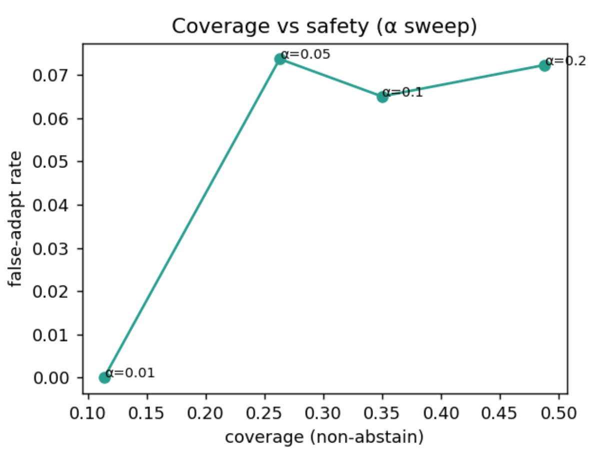


Fig. 7. **The safety knob.** Sweeping the level α trades decision coverage against the false-adapt rate: the certificate holds false-adapt at or below target while coverage grows with α, so a deployer chooses the operating point rather than accepting a fixed one.

regret and false-adapt audit. Stress grids use leave-one-cell-out residuals; semantic domain-shift tracks use a dev/test split. We report $\mathrm{FA_u} = \mathbb{P}[\textsc{adapt}, \Delta \le 0]$ and do not interpret $\mathrm{FA_c}$ as a certificate.

candidate	policy	mean acc	regret↓	harmful cells	beats both?
Tent	always-adapt	0.399	0.0216	63%	
	always-freeze	0.410	0.0114		
	K-Bound	**0.410**	**0.0114**		no (ties freeze)
EATA	always-adapt	**0.443**	0.0000	0%	no (ties adapt)
	always-freeze	0.410	0.0336		
	K-Bound	**0.443**	0.0000		
SAR	always-adapt	0.379	0.0625	15%	
	always-freeze	0.410	0.0319		
	K-Bound	**0.431**	**0.0108**		**yes**

TABLE XIII
IMAGENET-C, MECHANISM-FAITHFUL SAR (27 CELLS: 3 NOISE CORRUPTIONS, RESNET-50, α=0.1); FULL-SCALE RUN (2026-07-09) SUPERSEDING THE EARLIER 36-CELL PROVISIONAL CONFIGURATION.

a) Evidence tiers.: **Locked** means the table row traces to raw cells/seeds and a locked aggregate; **reconciled** means a prior pooled interpretation was corrected from its raw records; **provisional** means a derived out-of-fold summary exists but does not reproduce the nearby canonical scorer; **diagnostic** means a completed run that is deliberately not promoted. Only locked and reconciled rows support the empirical headline.

TABLE XIV

Uniform nine-track empirical panel. Regrets are ordered as KGA / always-adapt / always-freeze; lower is better. BB = beats both fixed policies. The final column is part of the result: it prevents a point estimate, a reconstructed summary, and a locked multi-seed result from being presented as equivalent evidence.

Track	Evaluation unit	Result	Decision interpretation	Evidence tier
CIFAR-10-C stress	5×432 cells / candidate	Tent: 0.0016/0.0079/0.1241; EATA: 0.0013/0.0033/0.1314; $\mathrm{FA_u} = 0$	Tent/EATA CI BB; SAR ties adapt	locked multi-seed
ImageNet-C SAR	27 cells (3 noise corr.)	0.0108/0.0625/0.0319; $\mathrm{FA_u} = 0$	**CI BB**; gap-freeze $[-.032, -.011]$, Holm; 12/27 abstain	locked single-seed full-scale
Camelyon17 OOD	genuine OOD test $n = 18$ (support $n = 36$)	0.0000/0.0000/0.1381; $\mathrm{FA_u} = 0$	ties adapt; no BB	reconciled raw records
iWildCam H v2	held-out test $n = 72$	0.0041/0.1028/0.0041; $\mathrm{FA_u} = 0$	no-harm, ties freeze	reconciled (OOF lock)
Office-Home M v2	target test $n = 35$	0.0157/0.0468/0.0158; $\mathrm{FA_u} = 0$	no-harm, tiny point edge	reconciled (OOF lock)
RxRx1 J	OOD test $n = 60$	0.0000/0.2531/0.0000; $\mathrm{FA_u} = 0$	ties freeze; no BB	locked raw run
PACS	four LODO targets, 108 cells each	three safety targets ($\mathrm{FA_u}$=0); one null (photo, $\mathrm{FA_u}$=0.056)	no promoted BB	locked single-seed panel
ImageNet-R D	3 seeds $\times 10$ backbones	no CI-robust BB; one point-only arm fails the second CI	abstention / fixed-policy ties	locked multi-seed diagnostic
CIFAR-10.1 K	cross-seed test $n = 48$	0.0021/0.0190/0.0017; $\mathrm{FA_u} = 0.444$	fails transfer bar; no claim	diagnostic; replay pending

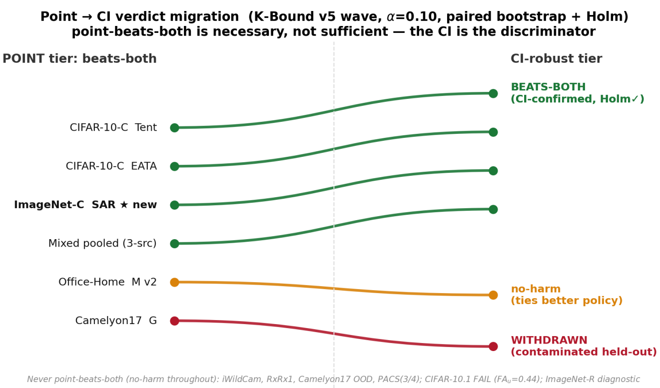


Fig. 8. **Point → CI verdict migration for the confirmatory wave** (α=0.10; paired per-condition bootstrap, Holm-corrected across the wave). Point-beats-both is necessary but not sufficient: under the CI, only CIFAR-10-C Tent/EATA, the constructed three-source mixture, and *ImageNet-C SAR* remain beats-both; Office-Home's point edge migrates to no-harm, and the contaminated Camelyon17 Protocol-G arm stays withdrawn. ImageNet-C SAR is the arm newly CI-confirmed in this fold-in pass. *Reproducibility:* verdicts from `scripts/uniform_scorer.py` + `scripts/percondition_bootstrap.py` (B=5000), scored once against the frozen bars in `results_source.json`.

TABLE XV

Primary numeric evidence used for empirical claims. The three-source row is a constructed routing mixture, not a natural-transfer result.

Track	Unit	KGA	Adapt	Freeze	Status
CIFAR-10-C Tent	5×432	.0016	.0079	.1241	CI BB, $\mathrm{FA_u} = 0$
CIFAR-10-C EATA	5×432	.0013	.0033	.1314	CI BB, $\mathrm{FA_u} = 0$
ImageNet-C SAR	27 cells	.0108	.0625	.0319	CI BB (Holm), $\mathrm{FA_u} = 0$
Camelyon17 OOD	$n = 18$	.0000	.0000	.1381	reconciled no-harm
RxRx1 J	$n = 60$	.0000	.2531	.0000	locked no-harm
Three-source OOF mixture	$n = 143$	.0059	.0632	.0342	CI BB (constructed); locked

G. Consolidated Claim and Guarantee Accounting

Statement	Status	Requirement
Abstention needed in matched-evidence opposite-benefit worlds	theorem	stated model class
$\mathrm{FA}_u \leq \alpha$ (unconditional false-adapt)	theorem	valid coverage + exchangeability + risk alignment
$\mathrm{FA}_c \leq \alpha$ (conditional false-adapt)	**not claimed**	report FA_c descriptively only
Nine-track panel	**accounting**	locked/reconciled/provisional tiers shown explicitly
Stress-grid beats-both (Tent/EATA)	confirm.	locked Protocol A v1 stress grid
ImageNet-C SAR CI-confirmed beats-both	confirm.	full-scale 27-cell noise grid; gap-freeze CI excl. 0, Holm-robust
Reconciled natural safety (Camelyon17, RxRx1)	confirm.	raw OOD/test artifacts
Office-Home and iWildCam OOF summaries	reconciled	OOF bootstrap lock
Three-source mixed routing aggregate	confirm.	constructed OOF mixture, not transfer
Universal improvement	**not claimed**	impossible in general
ε estimates drift budget β	**false**	β is declared; ε is empirical

VIII. Ablations and Sensitivity

A. Value of the Uncertainty Radius

Table X isolates the radius: KGA without the radius has lower point regret but a higher empirical false-adapt rate, whereas the full certificate sacrifices coverage to eliminate observed harmful commitments on the locked grid. This comparison separates benefit-prediction quality from certification.

B. Sensitivity Ablations

The ablations below reuse the locked CIFAR-10-C stress evidence—the logged per-condition Z and true benefit B (432 conditions/candidate, seed 0)—and the deployed rule (cross-fit gradient-boosted $\widehat{\Delta}$ plus conformal radius), so no target labels or model re-runs are involved (`scripts/ablation_sweep.py`). At the deployed operating point (Tent, α=0.10) the harness reproduces the locked gate row of Table X (regret 0.0017, $\mathrm{FA_u}$=0, adapt 0.51, coverage 0.68), confirming faithfulness.

(i) Safety level α. Sweeping α (Table XVI) trades decision coverage for regret while holding $\mathrm{FA_u}$=0 at *every* certified level; only the radius-free variant breaks the guarantee ($\mathrm{FA_u}$ = 0.051/0.032/0.007 for Tent/EATA/SAR). The certificate is a genuine safety knob, not one tuned point.

(ii) Estimator. Swapping the benefit regressor (Table XVII) leaves the false-adapt guarantee intact: gradient boosting (0.0017), random forest (0.0015), and ridge linear (0.0041) all keep $\mathrm{FA_u} \leq 0.002$; a small MLP over-abstains (regret 0.014, coverage 0.36) on this small calibration set. Safety is a property of the radius, not of the specific estimator.

(iii) Evidence family. Dropping any single evidence family (frozen, adapted, change, drift, or update) keeps regret within $[0.0013, 0.0019]$ at $\mathrm{FA_u}$=0: no single family is load-bearing, and the panel is deliberately redundant.

(iv) Cross-adapter transfer. Fitting the benefit model on one adapter and applying it to another (Table XVIII) *breaks* the false-adapt bound in most directions (SAR→Tent $\mathrm{FA_u}$ = 0.26, EATA→SAR 0.19, Tent→SAR 0.13): one gate does *not* transfer across adapters, which is exactly why the deployed method refits per adapter. Only Tent↔EATA transfer stays near-safe.

(v) Drift budget β. The population-frontier β sweep is the synthetic ground-truth validation of §VII-A ($\mathrm{FA_u} \leq 0.014$ across $\beta \in \{0.05, \dots, 0.20\}$), kept distinct from the empirical radius ε.

TABLE XVI
Ablation (i): safety-level α sweep, CIFAR-10-C Tent (n=432). $\mathrm{FA_u}$=0 at every certified α; only the radius-free row breaks it.

α	regret↓	$\mathrm{FA_u}$	adapt	coverage
0.01	0.0033	0.000	0.45	0.49
0.05	0.0020	0.000	0.49	0.64
0.10	0.0017	0.000	0.51	0.69
0.20	0.0011	0.000	0.53	0.74
no radius	0.0004	0.051	0.68	1.00

TABLE XVII
Ablation (ii): benefit estimator, CIFAR-10-C Tent (α=0.10). The radius keeps $\mathrm{FA_u} \approx 0$ regardless of estimator.

estimator	regret↓	$\mathrm{FA_u}$	adapt	coverage
gradient boosting	0.0017	0.000	0.51	0.69
random forest	0.0015	0.002	0.52	0.66
ridge (linear)	0.0041	0.000	0.46	0.48
small MLP	0.0143	0.000	0.36	0.36

IX. Discussion and Failure Modes

A. Why Natural Shifts Mainly Produce No-Harm

A single natural benchmark is often one-sided: adaptation helps almost everywhere or harms almost everywhere. In the first case, always-adapt is already near-oracle; in the second, always-freeze is near-oracle. A three-way router gains a strict margin over both fixed policies only when helpful and harmful conditions coexist and the evidence distinguishes them. Natural no-harm results and mixed-regime wins are therefore different operating points of the same decision geometry.

B. Four Sources of Abstention

Not every abstention has the same interpretation. *Structural non-identifiability* means opposite-benefit worlds share the same admissible evidence law. *Finite-sample uncertainty* means the sign is identifiable in principle but the interval still crosses zero. *Estimator inadequacy* means the chosen model fails to extract an available signal. *Calibration failure* means a development relationship does not transfer to deployment. The paper should diagnose these separately rather than labeling every null result "unknowable."

C. Operational Failure Modes

D. When to Deploy K-Bound

K-Bound is most valuable when harmful updates are plausible, the cost of a bad commit is high, a frozen fallback is available, and historical or structural information supports calibration. It is less useful when adaptation is uniformly benign, when no credible drift class or calibration set exists, or when the candidate update cannot be evaluated and rolled back safely.

X. Limitations

Scope of empirical wins. On the stress grid KGA beats both for Tent/EATA and ties SAR (Table IX); on ImageNet-C the pattern flips by candidate—Tent/EATA are robust while SAR collapses and KGA beats both under a mechanism-faithful re-run (Table XIII). Per-corruption CIFAR-10-C and helpful natural shifts are insurance regimes, not policy wins.

Real-shift scope. The uniform panel deliberately distinguishes three outcomes. Camelyon17 genuine OOD and

TABLE XVIII

Ablation (iv): cross-adapter transfer (α=0.10). Transferring one benefit model across adapters violates $FA_u \leq \alpha$ in most directions—per-adapter calibration is required.

fit → apply	regret↓	FA_u	adapt	coverage
Tent → EATA	0.0020	0.000	0.54	0.64
EATA → Tent	0.0010	0.007	0.55	0.69
Tent → SAR	0.0173	0.125	0.50	0.72
EATA → SAR	0.0236	0.192	0.57	0.63
SAR → Tent	0.0086	0.259	0.85	0.85
SAR → EATA	0.0028	0.123	0.76	0.76

TABLE XIX

Deployment failures, diagnostics, and safe responses.

Failure	Diagnostic	Safe response
Calibration transfer fails	support/residual audit	recalibrate, widen radius, or abstain
True drift exceeds β	external domain monitoring	revise class/budget; freeze meanwhile
Evidence out of support	distance or density check in Z-space	abstain
Adapter/checkpoint changed	configuration hash mismatch	require recalibration
Batch too small	minimum-size rule	wait for more data or freeze
Continual state accumulates error	checkpoint/stream monitor	rollback to last certified state
Missing/invalid evidence	schema and numerical validation	freeze

RxRx1 are reconciled/locked no-harm results: KGA ties the better fixed policy at zero observed false-adapt. Office-Home and iWildCam are reconciled from their out-of-fold bootstrap lock (`research_lock/KBOUND_WIN_BOOTSTRAP_CIS_oof.json`): no-harm safety results now backed by saved OOF confidence intervals rather than a pending replay. PACS, ImageNet-R, and CIFAR-10.1 are completed breadth tracks, not natural beats-both claims. The three-source mixture is a constructed routing aggregate; it is not a transfer result and does not change the per-dataset conclusion.

Theory and calibration. The conformal radius assumes exchangeable calibration pairs; sign bracketing without structure is impossible (Theorems 1 and 2; Appendix A). KGA's value scales with how often catastrophic, detectable harm occurs, not with universal accuracy gains. On helpful-dominated shifts the resulting near-*tie* with always-adapt reflects the certificate abstaining rather than committing harmful updates: KGA approximately matches the better policy, at most paying a small abstention cost when it declines a helpful but uncertifiable update, so the value proposition is conditional, like insurance against detectable harm. *Setting β.* The budget is *declared*, not estimated—a deployment-time upper bound on admissible calibration drift, fixed from domain knowledge, historical dev-to-deployment gaps, or an explicit stress envelope. The choice β=0 is the strongest zero-drift assumption and may be reported only as a plug-in baseline; it is not an assumption-free or conservative default. When no credible bound is justifiable, the population frontier cannot certify robustness to calibration drift, and the safe operational response is to abstain or freeze. Larger β widens the abstain band monotonically, trading decision coverage for robustness.

Detectability and transfer. ImageNet-R is a weak, split-specific evidence regime: one point-only arm does not survive both comparisons across three seeds. CIFAR-10.1 fails the cross-seed certificate bar (FA_u = 0.444). These are useful boundary cases, not failed headline experiments: they show that a low-margin signal should not be promoted merely because one split or one candidate looks favorable.

The limitations are the frontier, not flaws. The regime dependencies above are not three independent weaknesses but one phenomenon: the benefit-sign frontier (Theorem 2) seen from the operating side. Tying always-adapt on a helpful-dominated shift is the no-harm guarantee *holding*: one cannot beat an oracle that is already always-adapt. Declining ImageNet-R is the *unknowable* branch firing: the observable margin falls below the drift budget, so no label-free rule can decide the benefit sign, however good the adapter (Theorem 1). The coverage drop is mandatory abstention, not a tuning loss: committing in the unknowable region would be a false-adapt. None can be removed without violating the impossibility result, since a method that "won everywhere" would be claiming to decide the sign of an unidentifiable quantity. The one axis that genuinely improves all three is a *richer label-free evidence map*, which raises the observable margin $|M|$ so more shifts clear the budget (more knowable shifts, fewer abstentions, higher coverage), together with a *tighter radius* (jackknife+ or more calibration data) that reclaims coverage on knowable cells abstained only out of conservatism; both remain bounded by the frontier they cannot cross.

Baselines and independent validation. On the ImageNet-C SAR configuration, the committal confidence gate adapts on harmful cells while KGA abstains on the low-margin band. The POEM/AETTA comparison is complete only for protocol-matched ports; official implementations remain future baseline work. Independent replay is required for the Office-Home/iWildCam OOF summaries, and more seeds would strengthen ImageNet-C and PACS. Abstention lowers *decision* coverage but never withholds a prediction: it retains the frozen fallback f_0.

What we do not claim. Universal accuracy gains; adapter selection by test regret; multicandidate routes with violated false-adapt bounds; or wins on helpful-dominated shifts where always-adapt is already oracle-optimal.

XI. Reproducibility

A. Artifacts and Result Lineage

All locked/reconciled rows in Tables XIV–XV link to saved raw cells or seed aggregates in the anonymized repository. Provisional rows are retained only to document completed evidence and the required replay; they are not reused as headline claims. The core certificate module

TABLE XX

Claim → support status. Each headline claim and how it is backed: *proven* (formal theorem), *empirical* (pre-registered, scored once on held-out data), or *pending* (pre-registered but not yet run). Quantitative backing for the empirical rows is in the cited tables.

Claim	Support
Abstention necessary under matched evidence with opposite benefit (Thm 1)	proven
Benefit-sign frontier: identifiable iff $\|M\|>\beta$ (Thm 2)	proven
Finite-sample $\mathrm{FA_u}\leq\alpha$ under coverage (Thm 3)	proven (Lean-checked)
Nine-track evidence accounting	empirical (tiered)
CI beats-both on CIFAR-10-C stress (Tent/EATA)	empirical (locked)
CI-confirmed beats-both on ImageNet-C SAR	empirical (27-cell full-scale noise grid)
Reconciled natural no-harm on Camelyon17 OOD and RxRx1	empirical (reconciled/locked)
Mixed head-to-head WIN vs. POEM/AETTA ports (Table XI)	empirical (pre-reg.)
Three-source constructed mixture beats both	empirical (locked; not transfer)
Single-dataset natural-shift beats-both	not claimed
Universal accuracy improvement	not claimed

is packaged as an installable module in the anonymized repository.

B. Formal Verification Scope

A kernel-checked Lean 4 formalization of the *algebraic and finite-testing core* of K-Bound accompanies the repository (43 indexed theorems, no `sorry`/`admit`/project-defined axioms). The certificate's probability layer is machine-checked as well: coverage implies the unconditional false-adapt/false-freeze bound over an *arbitrary probability measure* (`measure_false_adapt_le_alpha`), and split-conformal coverage holds in the conditional-on-the-bag uniform-rank model (`uniformIndex_false_adapt_le`); we are not aware of a prior machine-checked conformal-coverage guarantee in a proof assistant (kernel-checked generalization bounds exist, e.g. [75], but not conformal coverage). General exchangeable-process coverage, and the remaining deployment-facing assumptions (calibration transfer and risk alignment) remain explicit assumptions outside the formalization.

C. Claim-to-Support Map

XII. Conclusion

K-Bound reframes test-time adaptation as a validity decision rather than an unconditional update. The central object is the target benefit $\Delta = R_T(f_0) - R_T(f_a)$, and the central limit is identifiability: over a declared drift-budget class, the benefit sign is recoverable only when the observable disagreement-region margin dominates admitted calibration drift. When the evidence cannot distinguish opposite-benefit worlds, abstention is a required action rather than a heuristic preference.

KGA operationalizes this principle with a label-free benefit estimate and a pre-calibrated uncertainty radius. The resulting controller commits the candidate only when the interval excludes zero, retains the frozen model when harm is certified, and otherwise abstains from changing state. Across the benchmark panel, the observed behavior follows the regime map: controlled mixed regimes provide beats-both routing results, one-sided natural shifts primarily provide no-harm behavior, and weak or non-transferable evidence remains diagnostic rather than being promoted as a win.

The broader implication is that an adaptation system should expose two components: an update mechanism and a validity layer that decides whether the update is supportable under the available evidence and assumptions. Strengthening K-Bound therefore requires not only better adapters, but also richer risk-aligned evidence, more defensible calibration across shifts, explicit drift-budget auditing, and reproducible deployment semantics.

TABLE XXI

Base 11-dim evidence panel (stress grids).

Feature	Definition	Family
`pre_entropy`	$\mathrm{mean}_n H(p_{0,n})$	frozen output
`pre_conf`	$\mathrm{mean}_n \max_c p_{0,n}$	frozen output
`pre_pbal`	$H(\bar{m}_0)/\log C$	frozen marginal
`post_entropy`	$\mathrm{mean}_n H(p_{a,n})$	adapted output
`post_conf`	$\mathrm{mean}_n \max_c p_{a,n}$	adapted output
`post_pbal`	$H(\bar{m}_a)/\log C$	adapted marginal
`pbal_drop`	`pre_pbal`−`post_pbal`	change
`entropy_drop`	`pre_entropy`−`post_entropy`	change
`frac_highconf`	$\mathrm{frac}_n(\max_c p_{a,n} > 0.9)$	collapse
`marginal_KL`	$\sum_c \bar{m}_{a,c} \log(\bar{m}_{a,c}/\bar{m}_{0,c})$	drift
`update_norm`	$\|\Delta\theta_{\mathrm{aff}}\|_2$	update

Appendix

Algorithm 2 gives the shared evaluation protocol behind the uniform nine-track panel (Table XIV) and the primary numeric evidence (Table XV).

Two label-free evidence panels are used, matched to the two calibration regimes of §V. Notation: p_0, p_a are the frozen and adapted softmax rows on the adaptation batch; $\bar{m}_0 = \mathrm{mean}_n\, p_{0,n}$ and $\bar{m}_a$ are the predicted-class marginals; $H(p) = -\sum_c p_c \log p_c$; C is the number of classes. The stress-grid tracks (CIFAR-10-C, ImageNet-C, PACS) use the 11-dimensional base panel of Table XXI (`kbound_pkg/kbound/evidence.py`). The natural-shift tracks (Camelyon17, iWildCam, Office-Home, RxRx1) use the 16-dimensional panel of Table XXII: a 10-statistic frozen-logit summary plus a 6-feature rich panel (`scripts/run_wilds_camelyon17.py`). All features are target-label-free; the ATC feature calibrates its threshold on *source* labels only.

This appendix is not required for the main empirical claim; it preserves the full theorem stack and validator

Algorithm 2 Calibration and held-out evaluation for K-Bound

Require: Conditions $\mathcal{C}$; frozen f_0; adapters $\mathcal{A}_1, \dots, \mathcal{A}_K$; calibration split $\mathcal{C}_{\text{cal}}$; test split $\mathcal{C}_{\text{test}}$; level α

Ensure: Regret-to-oracle, false-adapt rate, coverage, beats-both verdict

for all $c \in \mathcal{C}_{\text{cal}}$ **do**
  for all adapters $\mathcal{A}_k$ **do**
    $f_a^{(k,c)} \leftarrow \mathcal{A}_k(f_0, X_c)$
    $Z_{k,c} \leftarrow \phi(X_c, f_0, f_a^{(k,c)})$
    $\Delta_{k,c} \leftarrow R_c(f_0) - R_c(f_a^{(k,c)})$
  end for
end for
Calibrate without test leakage: on stress grids, form leave-one-cell-out residuals $\rho_i = |\widehat{\Delta}_{-i}(Z_i) - \Delta_i|$; on semantic shifts, reserve a development calibration split disjoint from the declared test split
$\varepsilon \leftarrow (1-\alpha)$ empirical quantile of the calibration residuals
Fit deployment estimator $\widehat{\Delta}(Z)$ on *all* calibration pairs ▷ scores the held-out test split
for all $c \in \mathcal{C}_{\text{test}}$ **do**
  for all adapters $\mathcal{A}_k$ **do**
    Run Algorithm 1 with $(f_0, \mathcal{A}_k, X_c, \widehat{\Delta}, \varepsilon)$
    Record decision, regret-to-oracle, coverage, false-adapt indicator
  end for
end for
Aggregate metrics vs. always-adapt, always-freeze, per-condition oracle

TABLE XXII
RICH 16-DIM PANEL (NATURAL SHIFT): 10 FROZEN-LOGIT STATISTICS + 6 RICH FEATURES. H IS SOFTMAX ENTROPY, $s = \max_c \text{softmax}$ THE CONFIDENCE, $E(x) = -\log \sum_c e^{z_c}$ THE FREE ENERGY; SUBSCRIPTS s, t DENOTE SOURCE/TARGET POOLS.

Feature	Definition	Family
Frozen-logit summary (10)		
ent mean/std	mean/std H	frozen
ent p25/p75	25/75 pct. of H	frozen
conf mean/std	mean/std s	frozen
frac hi-conf	frac(s>0.9)	frozen
frac hi-ent	frac(H> log 2)	frozen
marg. max	$\max_c$ mean_n p	frozen marginal
marg. std	$\text{std}_c \text{mean}_n$ p	frozen marginal
Rich features (6)		
`disagree`	frac($\arg\max f_0 \neq \arg\max f_a$)	disagreement
`ent_gap`	mean $H(f_0)$ − mean $H(f_a)$	change
`en_shift`	mean E_t − mean E_s	drift
`bn_kl`	mean_{ch} $\text{KL}(\mathcal{N}_{\text{run}} \Vert \mathcal{N}_{\text{batch}})$	drift
`atc`	$\text{frac}_t(s \geq t)$, t src-calib.	accuracy proxy
`conf_drop`	mean s_s − mean s_t	drift

mapping behind the three main-paper theorems: matched-evidence impossibility (Theorem 1), the benefit-sign frontier (Theorem 2), and the finite-sample certificate (Theorem 3). Below we give the full five-theorem stack, lemmas, and proof sketches with validator references.

TABLE XXIII
PER-TRACK ADAPTER HYPERPARAMETERS (FROM THE RUN CONFIGS). ADAPTED PARAMETERS ARE BN/LAYERNORM-AFFINE ONLY; TTA OPTIMIZER IS ADAM. "AGGRESSIVENESS" CELLS USE MILD = (10 STEPS, LR 10^{-3}) OR AGGRESSIVE = (50 STEPS, LR 2.5×10^{-3}). EATA USES AN ENTROPY FILTER e_{thr}=0.4 ln 2≈0.28; SAR ADDS SAM + ENTROPY RESET. BACKBONES FOR CIFAR-10-C, IMAGENET-C, PACS, AND CAMELYON17 ARE VERIFIED FROM THE RUN CODE; THE REMAINING WILDS/DOMAINBED TRACKS USE THE STANDARD BASELINE.

Track	Backbone	Adapter(s)	Steps / lr
CIFAR-10-C stress	ResNet-18	Tent/EATA/SAR	mild & aggressive (per cell)
ImageNet-C	ResNet-50	Tent/EATA/SAR (SAM+reset)	mild & aggressive (per cell)
PACS (LODO)	ResNet-18	mild/aggressive	$10/10^{-3}$ & $50/2.5\times10^{-3}$
Camelyon17	DenseNet-121	EATA online	10 / 10^{-3}
iWildCam	ResNet-50	Tent episodic	10 / 10^{-3}
Office-Home	ResNet (DomainBed)	SAR online aggr.	50 / 2.5×10^{-3}
RxRx1	ResNet-50	episodic	5 / (batch 64)

a) Extensions beyond the main-text spine.: The lemmas and theorems below strengthen the three load-bearing results (Lemmas 1, 1; Theorems 2, 3) with minimax floors, one-bit structure, and rate statements. Full proofs and validators are in `kbound.tex` and `theory_v2/`.

Lemma 2 (Plug-in regret identity)**.** *For any estimator $\hat{\Delta}$ and gate $\hat{g} = \mathbf{1}[\hat{\Delta} > 0]$,*

$$R(\hat{g}) - R(g^\star) = \mathbb{E}\big[|\Delta(Z)|\, \mathbf{1}\{\text{sign}\,\hat{\Delta} \neq \text{sign}\,\Delta\}\big], \qquad g^\star = \mathbf{1}[\Delta > 0].$$

On $\{|\Delta| < \varepsilon\}$, committal regret is at most ε.

Proof. Wrong-sign commits pay $|\Delta|$ pointwise; averaging gives the identity. ☐

Proposition 1 (Finite-sample minimax regret floor)**.** *Let two target worlds share evidence laws at TV distance δ and have opposite benefit signs with magnitude $\Lambda > 0$. For every label-free committal gate and finite n,*

$$\inf_{\hat{g}} \max_{\theta = \pm 1} \big[R_{P_\theta}(\hat{g}) - R_{P_\theta}(g^\star)\big] \;\geq\; \tfrac{\Lambda}{2}(1-\delta) \;\geq\; \tfrac{\Lambda}{4}\, e^{-\text{KL}}.$$

A Gaussian family with $d_n = c/\sqrt{n}$ realizes a constant floor in n (`val_thm2_lecam_finite_n.py`).

Theorem 4 (Matched evidence impossibility (full form))**.** *Theorem 1 is part (iii) of the following. Fix $\alpha \in (0, \frac{1}{2})$.*

(i) *Witness worlds with $\text{Law}(Z \mid P_T^1) = \text{Law}(Z \mid P_T^2)$ and $\text{sign}\,\Delta_1 = -\,\text{sign}\,\Delta_2 \neq 0$.*
(ii) *Le Cam identity: $\inf_g M(g) = 1 - \text{TV}$ between the evidence laws.*
(iii) *Forced abstention at rate $\geq 1 - 2\alpha$ under dual error control.*

Corollary 2 (Abstention is mandatory, not conservative)**.** *In the $|M| \leq \beta$ wedge of Theorem 2, Theorem 4(iii) implies any label-free rule with false-adapt and false-freeze $\leq \alpha$ must abstain with probability at least $1 - 2\alpha$ on matched evidence.*

Proposition 2 (Certificate sample complexity)**.** *With empirical-Bernstein paired benefits, one-sided certification requires $n = \tilde{O}(\sigma^2/\Delta^2)$; two-sided sign certification requires the same exponent with a larger constant (`val_minimax_optimality.py`).*

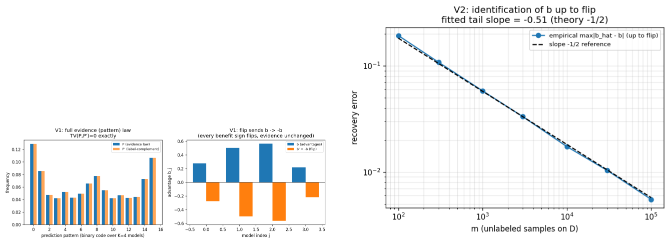


Fig. 9. One-bit identification (Synthetic validator; Thm. 5(ii)).

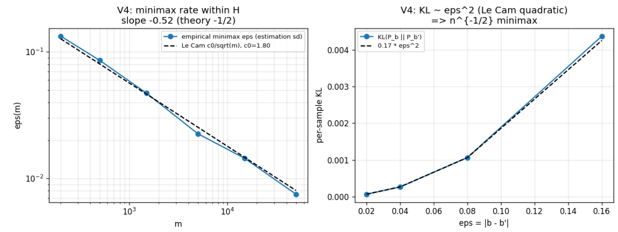


Fig. 11. Evidence-channel rate (Synthetic validator; Thm. 6).

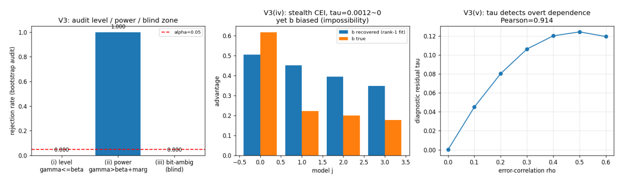


Fig. 10. Budget audit (Synthetic validator; Thm. 5(iv)).

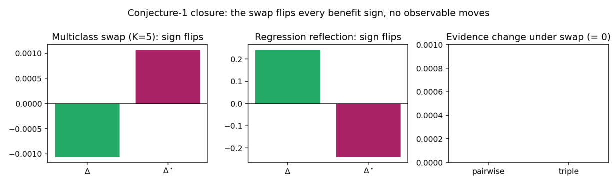


Fig. 12. Conjecture 1 closure validator.

Remark 3 ($\beta = 0$ face). ATC, DoC, GDE, COT, AETTA, and agreement-on-the-line are the $\beta = 0$ face of Theorem 2: sound iff $\gamma = 0$.

Theorem 5 (One-bit dichotomy)**.** *Under the CEI panel hypothesis* $\boldsymbol{H}$*:* (i) *rank-one agreement law;* (ii) *evidence fixes magnitudes but not global orientation* $b \mapsto -b$*;* (iii) *no evidence-definable assumption identifies* sign Δ*; the minimal untestable supplement is one bit;* (iv) *drift budgets are auditable but not verifiable on a blind set. Conjecture 1 is resolved negatively by (iii).*

Conjecture 1 (Label-free bracketing — **resolved**)**.** *Unconditional label-free bracketing does not exist; weakest one-bit classes are in Appendix B (Theorems 7, 8).*

Theorem 6 (Evidence-channel rate)**.** *Under audited H, unlabeled agreement statistics and labeled paired benefits both converge at rate $m^{-1/2}$; labels change constants and supply the missing bit, not the exponent.*

Validator map. `val_thm2_lecam_finite_n.py` (Prop. 1); `val_conj1_*.py` (Thm. 5); `val_ev_rate.py` (Thm. 6).

A. Validation figures

The main text contains the complete empirical accounting in the uniform nine-track panel. Historical route studies, superseded per-dataset tables, and outcome-free camera templates are deliberately excluded from this short-paper build. Their raw logs remain in the repository for audit, but they are not independent empirical claims. The declared evaluation unit is a condition or held-out cell, never an arbitrary pool of frames or domains; all policies replay the same unit stream, and labels are revealed only after the KGA decision for offline scoring.

The theory-validation suite supplies numerical sanity checks. Multi-seed experiments use pooled means with paired bootstrap where this was pre-registered; a point estimate or a single locked configuration is labeled as such in Table XIV.

B. Weakest one-bit class

The weakest one-bit class (resolving the open piece of Conjecture 1). Theorem 5 fixes the supplement at *one bit*; what remained open is the *weakest falsifiable class* on which that bit suffices. We settle it conditionally. On D write $a(x) := 2\eta_a(x) - 1$ with $\eta_a(x) = \mathbb{P}_T(f_a{=}Y \mid x)$, so $\Delta = \int_D a\, d\mu$ (Lemma 1). Let $m(x) := 2s(x) - 1$ be the observable relative (calibrated-score) margin, so $M = \frac{1}{2}\mathbb{E}_{\mu|D}[m]$, and let the observable *relative-calibration field* be $c(x) := w(x)\,(m(x) - m^\star)$ with observable nonnegative weight $w(x)$ and observable neutral anchor $m^\star$ (the margin at which $s = \frac{1}{2}$). Put $T_E := \int_D c\, d\mu$ (observable), and let E denote the law of all label-free observables; two targets are *evidence-identical* when they share E (equivalently (μ, m, s), since predictions and s are fixed maps of x).

Definition 3 (Margin-monotone relative-calibration class)**.** $\mathcal{C}_{\text{mono}}$ is the class of targets for which $a(x) = \sigma\, c(x)$ μ-a.e. on D for a single global orientation $\sigma \in \{\pm 1\}$; equivalently the benefit is single-crossing in the observable margin at $m^\star$ (sign $a(x) = \sigma$ sign$(m(x) - m^\star)$) with calibrated magnitude $|a(x)| = |c(x)|$. The class is *falsifiable but not verifiable*: a finite labelled probe can reject "$a = \pm c$", yet σ is never label-free identified.

Assumption 2 (General position)**.** Within an evidence fibre the regions of D whose benefit sign is free carry comparable, not-fully-E-pinned benefit mass: $T_E \neq 0$ and no proper subcollection of free regions has an E-certifiable dominant total $\int |a|\, d\mu$.

Lemma 3 (Structural sign-freedom = bit complexity)**.** *Fix E and a falsifiable class $\mathcal{C}$, and let $K(\mathcal{C}, E)$ be the number of disjoint μ-positive regions of D whose benefit signs are jointly free in $\mathcal{C}$ on the fibre E. Under Assumption 2, a* *falsifiable structural supplement of b bits pins* $\operatorname{sign}\Delta$ *across the fibre* iff $b \geq K(\mathcal{C}, E)$.

Proof. If $b \geq K$, declaring the orientation of each free region fixes the sign field; the rest of a being E-pinned, $\Delta = \int_D a\, d\mu$ then has E-computable sign. If $b < K$, some free region R is undeclared; the two targets agreeing off R and differing in $\operatorname{sign} a$ on R are evidence-identical and share all b declared bits, while by comparability of $\int_R |a|$ with the complement (Assumption 2) their benefits have opposite sign—no decoder of the b declared bits separates them. □

Theorem 7 (Conditional resolution of Conjecture 1: a weakest one-bit class)**.** (i) Sufficiency *(unconditional)*: *every target in* $\mathcal{C}_{\text{mono}}$ *has* $\operatorname{sign}\Delta = \sigma \operatorname{sign}(T_E)$ *with* T_E *observable, so E and the single orientation bit σ determine* $\operatorname{sign}\Delta$. (ii) Necessity *(unconditional)*: *σ and $-\sigma$ are evidence-identical (*$\operatorname{Law}(E \mid \sigma) = \operatorname{Law}(E \mid -\sigma)$*) yet give Δ and $-\Delta$, so by Le Cam at least one bit is required. Thus* exactly one bit certifies $\operatorname{sign}\Delta$ on $\mathcal{C}_{\text{mono}}$, with no further assumption. (iii) Minimality *(under Assumption 2)*: *every falsifiable $\mathcal{C} \supsetneq \mathcal{C}_{\text{mono}}$ contains a non-margin-monotone target with $K(\mathcal{C}, E) \geq 2$, so by Lemma 3 at least two bits are required. Hence $\mathcal{C}_{\text{mono}}$ is* a *weakest falsifiable one-bit class—minimal in the single-crossing, calibrated-magnitude direction; it is not unique (Remark 4).*

Proof. (i) $\Delta = \int_D a\, d\mu = \sigma \int_D c\, d\mu = \sigma T_E$. (ii) Sending $\sigma \mapsto -\sigma$ is the D-local label complement of Theorem 5: it fixes (μ, m, s), hence E, while negating a and thus Δ; the two-point bound of Theorem 4 gives minimax committal error $\frac{1}{2}$. (iii) A target outside $\mathcal{C}_{\text{mono}}$ violates $\operatorname{sign} a = \sigma \operatorname{sign}(m - m^\star)$ on a μ-positive region, so its sign field is not one global orientation; in the swap-closed (falsifiable) fibre at least two regions are independently signable, i.e. $K \geq 2$, and Lemma 3 applies. □

Proposition 3 (Explicit non-identifiability witness)**.** *For each candidate single structural bit there are two targets with identical evidence law (*$\mathrm{TV} = 0$*) sharing that bit's value but with opposite* $\operatorname{sign}\Delta$. *Take $D = R_1 \cup R_2$, $\mu(R_1) = \mu(R_2) = \frac{1}{2}$, with R_1 calibrated (c_1=1) and R_2 at the anchor (c_2=0, no observable benefit signal). For $\beta = \operatorname{sign} a_1$: $a = (+1, -2)$ vs. $(+1, +2)$ give $\Delta = -\frac{1}{2}$ vs. $+\frac{3}{2}$. For $\beta = \operatorname{sign} a_2$: $a = (+1, +\frac{1}{5})$ vs. $(-1, +\frac{1}{5})$ give $\Delta = +\frac{3}{5}$ vs. $-\frac{2}{5}$. No single bit certifies* $\operatorname{sign}\Delta$*, so two bits are necessary off $\mathcal{C}_{\text{mono}}$.*

Remark 4 (What is unconditional, and what rests on general position). Parts (i)–(ii)—that one declared bit certifies $\operatorname{sign}\Delta$ on $\mathcal{C}_{\text{mono}}$—hold *unconditionally.* Only minimality (iii) uses Assumption 2: if it fails (one region's benefit mass is E-certifiably dominant) that region's orientation alone certifies $\operatorname{sign}\Delta$ even outside $\mathcal{C}_{\text{mono}}$, moving the boundary. Equivalently, Assumption 2 is the requirement that the certificate be *domination-free* (minimax-sound over the unobserved magnitudes); so the General-Position statement is only *conditional*; the unconditional weakest classes are characterized in Theorem 8 as an explicit finite family of dominance polytopes. Even under Assumption 2, $\mathcal{C}_{\text{mono}}$ is only *a* weakest one-bit class: any calibrated sign-aligned field $\{a = \sigma h : h \text{ observable}\}$ is equally one-bit and is incomparable to $\mathcal{C}_{\text{mono}}$. All six checks are machine-checked by a deterministic release validator: the unconditional core (sufficiency/necessity 20000/20000, $|\Delta - \sigma T_E| = 0$ exactly), the explicit minimality witnesses of (iii), the general-position ablation, and the non-uniqueness check.

Why general position cannot simply be dropped (the precise obstruction, and exactly what is open). The minimal counterexample is two equal-mass regions $D = R_0 \cup R_1$, $\mu(R_0) = \mu(R_1) = \frac{1}{2}$, with the observable field $c = +1$ on the *calibrated* region R_0 and $c = 0$ on the *anchor* region R_1 (no observable benefit signal there, so R_1 is sign-free in any evidence fibre). Enlarge $\mathcal{C}_{\text{mono}}$ to the falsifiable class $\mathcal{C}_{\text{dom}} = \{a_{R_0} = \sigma,\ |a_{R_1}| \leq \rho\}$ for a declared constant $\rho < 1$. Since $\Delta = \frac{1}{2}(a_{R_0} + a_{R_1})$ and $|a_{R_1}| \leq \rho < 1 = |a_{R_0}|$, the calibrated region dominates and $\operatorname{sign}\Delta = \sigma$ for *every* member, so one declared bit certifies $\operatorname{sign}\Delta$ on all of $\mathcal{C}_{\text{dom}}$. Yet $\mathcal{C}_{\text{dom}} \supsetneq \mathcal{C}_{\text{mono}}$ (it contains non-margin-monotone targets with $a_{R_1} \neq 0$, e.g. $a = (+1, \pm\frac{1}{2})$ giving $\Delta = \frac{3}{4}, \frac{1}{4}$, both positive) and is falsifiable (a labelled probe rejects targets that are uncalibrated on R_0 or violate $|a_{R_1}| \leq \rho$). This *contradicts* minimality once an E-certifiable dominant region is permitted: dropping $a = (+1, \pm 2) \Rightarrow \Delta = +\frac{3}{2}, -\frac{1}{2}$ shows that the moment $\rho \geq 1$ the anchor region can flip $\operatorname{sign}\Delta$ and two bits become necessary (K=2). The obstruction to an *unconditional* weakest class is therefore exactly this: characterising minimality across the *entire* lattice of dominant-region thresholds ρ (where the weakest one-bit class is not $\mathcal{C}_{\text{mono}}$ but a ρ-dependent family), which Assumption 2 sidesteps by fiat. Theorem 8 (below) resolves it: the ρ-lattice is the slice $\{r_0{=}1,\ r_1{=}\rho\}$ of the single dominance polytope $W^\star{=}\{r_0 \geq r_1\}$, with ρ=1 its boundary. The construction is machine-checked (`val_conj1_genpos.py`, seeds fixed): $\mathcal{C}_{\text{dom}}$ strictly contains $\mathcal{C}_{\text{mono}}$, is falsifiable (reject rate 1.0), is one-bit-certified (error rate 0 over 2×10^5 targets), and loses one-bit certifiability without the domination bound (error rate $\approx \frac{1}{4}$).

Unconditional resolution (General Position removed). Dropping Assumption 2 reveals, for each orientation pattern, a single maximal one-bit class—a *dominance polytope*—of which the ρ-family of Remark 4 is one slice.

Definition 4 (Orientation pattern and canonical class)**.** On a fibre E an *orientation pattern* is a tied set $P \subseteq \{i : c_i \neq 0\}$ with a tied sign-pattern $s \in \{\pm\}^P$; the free set is $Z = D \setminus P$. For an observable magnitude region $W \subseteq [0,1]^{|D|}$ the *canonical class* is $\mathcal{C}(P, s, W) = \{a : \operatorname{sign} a_i = \sigma\, s_i \operatorname{sign} c_i\ (i \in P),\ (|a_j|)_j \in W\}$ with one declared bit σ. Write $T(r) = \sum_{i \in P} s_i \operatorname{sign}(c_i) \mu_i r_i$ and $G(r) = \sum_{i \in Z} \mu_i r_i$.

Lemma 4 (Canonical form of a falsifiable class)**.** *A class is falsifiable* iff *it equals some $\mathcal{C}(P, s, W)$; anchors ($c_i = 0$)*

lie necessarily in Z. (Swap-closure, Thm. 5(iii), forces the constraint on Z to depend only on $|a|$; forcing an anchor sign is not evidence-definable. Both aligned ($s_i{=}+$) and anti-aligned ($s_i{=}-$) ties are falsifiable since σ is unidentified.)

Theorem 8 (Unconditional one-bit criterion; weakest classes)**.** *Let $\mathcal{C}(P, s, W)$ be falsifiable.* (i) (Dominance margin.) *One declared bit certifies* sign Δ *on $\mathcal{C}(P, s, W)$* iff $\inf_{r\in W}\big(T(r) - G(r)\big) \geq 0$ *(with some member $\Delta > 0$) or* $\sup_{r\in W}\big(T(r) + G(r)\big) \leq 0$. (ii) (Weakest class.) *For a fixed pattern (P, s) the unique maximal one-bit class is the dominance polytope $\mathcal{C}^\star = \mathcal{C}(P, s, W^\star)$, $W^\star = \{r \in [0, 1]^{|D|} : T(r) \geq G(r)\}$; $\mathcal{C}_{\rm mono}$, every $\mathcal{C}_{\rm dom}(\rho \leq 1)$, and every box one-bit class are proper subclasses of one $\mathcal{C}^\star$.* (iii) (Finite family.) *The set of weakest falsifiable one-bit classes is the explicit finite family $\{\mathcal{C}^\star\}$ indexed by $(P \subseteq \{c \neq 0\}$, $s \in \{\pm\}^P)$ modulo the global flip $\sigma \mapsto -\sigma$ (at most $3^{|\{c\neq 0\}|}$ patterns); distinct patterns give pairwise incomparable maxima, so there is no* unique *weakest class. $\mathcal{C}_{\rm mono}$ is the all-tied all-aligned member, and Assumption 2 is exactly the face on which the family collapses to $\mathcal{C}_{\rm mono}$, recovering Thm. 7(iii).*

Proof. Δ is linear, so over $\mathcal{C}(P, s, W)$ at bit $\sigma = +1$ it ranges over $[\inf_W(T - G), \sup_W(T + G)]$ (free signs independent by swap-closure; the $\sigma = -1$ fibre is its negation); a single bit certifies iff each fibre's strict sign set is a singleton, giving (i). For $W^\star$, $\inf_{W^\star}(T - G) \geq 0$ by construction, and any r_0 with $T(r_0) < G(r_0)$ admits, with opposing free signs, a member with $\Delta < 0$ while $W^\star$ admits $\Delta > 0$, so $W^\star$ is maximal; $\mathcal{C}_{\rm dom}(\rho)$ sits in $W^\star$ iff $\rho \leq 1$, with $\rho = 1$ the boundary, giving (ii). Incomparability (iii) is witnessed on $c = (1, 1, 0)$ by $a = (0, \frac{1}{3}, 0)$ and $a = (1, -\frac{1}{3}, \frac{1}{3})$. On the general-position face the free block can overpower the tied block unless Z-magnitudes vanish, leaving only $\mathcal{C}_{\rm mono}$. □

Verification. The criterion (i) was checked against exact brute-force vertex truth on $2.8{\times}10^5$ box fibres and $3.2{\times}10^3$ non-box integer polytopes (exact-LP inf / sup) with 0 mismatches; sufficiency on $W^\star$ over $1.2{\times}10^5$ members (0 errors), necessity just outside $W^\star$ (error rate $\approx \frac{1}{2}$), and the $\mathcal{C}_{\rm dom}(\rho)$ collapse ($\rho{\leq}1$ one-bit, $\rho{>}1$ lost) are machine-checked (`val_unconditional_weakest.py`, seeds fixed; fails loudly). The single non-tautological input is Lemma 4, which rests on the swap involution (Thm. 5(iii)).

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